



ICM-40609-D

A TDK SmartMotion™ Solution Designed for Drone Market

GENERAL DESCRIPTION

The ICM-40609-D is a 6-axis MEMS MotionTracking™ device that combines a 3-axis gyroscope and a 3-axis accelerometer designed for the drone market. The ICM-40609-D is form factor compatible with our legacy drone devices – making the transition to the latest offering incredibly simple.

The ICM-40609-D takes advantage of a state-of-the-art architecture design that enhances the IMU's performance and accuracy over temperature, making it the ideal drone solution to control the stability of the platform during long flight times that may experience high temperature shifts.

The ICM-40609-D has a max ODR of 32 KHz, making it the best sampling rate available in a consumer device. The ability to capture data at such a high ODR allows for customers to easily find any anomalies or errors that need to be addressed during flight. The accel Full Scale Range has also been increased to 32g, allowing for substantial linear movement to be easily tracked.

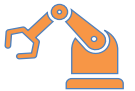
Other industry-leading features include on-chip 16-bit ADCs, programmable digital filters, an embedded temperature sensor, and programmable interrupts. The device features I²C and SPI serial interfaces, a VDD operating range of 1.71V to 3.6V, and a separate VDDIO operating range of 1.71V to 3.6V.

FEATURES

- Gyro Noise: **4.5mdps/√Hz**
- Gyro Offset Stability TC: **±10mdps/C**
- Gyro Sensitivity Error: **±0.5%**
- Gyro Sensitivity/temp: **±0.045%/C**
- Accel Noise: **100μg/√Hz**
- Accel Offset Stability TC: **±0.15mg/C**
- Accel Sensitivity Error: **±0.5%**
- Accel Sensitivity/temp: **±0.007%/C**
- Gyro + Accel Combo current: **0.77mA**
- Extended Accel Full Scale Range: **32g**
- Improved ODR Latency: **32KHz**

Custom architecture for Improved Thermal Gradient Behavior

- Best-in-class accuracy over temperature



Increased ODR/FSR for max data collection

- 32g Accel Full Scale Range
- 32 KHz ODR Sample Rate



Form Factor Compatible with Legacy Products

- Easily transition from ICM-20602 and MPU-6500
- Only minor pinout changes required



APPLICATIONS

- Drones, Flight Controller

PART NUMBER	PACKAGE SIZE	TARGET MARKETS	FULL SCALE RANGE	ODR & SAMPLE SYNCH	STATUS
MPU-6500	3x3x0.9mm 24-pin QFN	Various	±2000dps/16g	G: 8KHz/ A: 4KHz 16-bit	NR/ND
ICM-20602	3x3x0.75mm 16-pin LGA	Various	±2000dps/16g	G: 8KHz/ A: 4KHz 16-bit	NR/ND
ICM-42688-P	2.5x3 14mm-pin LGA	Robotics/HMD/loT/Drones	±2000dps/16g	32KHz G: 19-bit / A: 18-bit	Active
ICM-40609-D	3x3x0.91mm 24-pin LGA	Enhanced Drone Performance	±2000dps/32g	32KHz 16-bit	2H 2022

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1 INTRODUCTION

1.1 PURPOSE AND SCOPE

This document is a product specification, providing a description, specifications, and design related information on the ICM-40609-D Single-Interface MotionTracking device. The device is housed in a small 3x3x0.91 mm 24-pin LGA package.

1.2 PRODUCT OVERVIEW

The ICM-40609-D is a 6-axis MotionTracking device that combines a 3-axis gyroscope, and a 3-axis accelerometer in a small 3x3x0.91 mm (24-pin LGA) package. It also features a 2 KB FIFO that can lower the traffic on the serial bus interface and reduce power consumption by allowing the system processor to burst read sensor data and then go into a low-power mode. ICM-40609-D, with its 6-axis integration, enables manufacturers to eliminate the costly and complex selection, qualification, and system level integration of discrete devices, guaranteeing optimal motion performance for consumers.

The gyroscope supports eight programmable full-scale range settings from ± 15.625 dps to ± 2000 dps, and the accelerometer supports four programmable full-scale range settings from $\pm 4g$ to $\pm 32g$.

Other industry-leading features include on-chip 16-bit ADCs, programmable digital filters, an embedded temperature sensor, and programmable interrupts. The device features I²C and SPI serial interfaces, a VDD operating range of 1.71V to 3.6V, and a separate VDDIO operating range of 1.71V to 3.6V.

The host interface can be configured to support I²C slave or SPI slave modes. The I²C interface supports speeds up to 1 MHz, and the SPI interface supports speeds up to 24 MHz.

By leveraging its patented and volume-proven CMOS-MEMS fabrication platform, which integrates MEMS wafers with companion CMOS electronics through wafer-level bonding, InvenSense has driven the package size down to a footprint and thickness of 3x3x0.91 mm (24-pin LGA), to provide a very small yet high-performance low-cost package. The device provides high robustness by supporting 20,000g shock reliability.

1.3 APPLICATIONS

- Drones
- Robotics
- IoT Applications

2 FEATURES

2.1 GYROSCOPE FEATURES

The triple-axis MEMS gyroscope in the ICM-40609-D includes a wide range of features:

- Digital-output X-, Y-, and Z-axis angular rate sensors (gyroscopes) with programmable full-scale range of ± 15.625 , ± 31.25 , ± 62.5 , ± 125 , ± 250 , ± 500 , ± 1000 , and ± 2000 degrees/sec
- Low Noise (LN) power mode support
- Digitally programmable low-pass filters
- Factory calibrated sensitivity scale factor
- Self-test

2.2 ACCELEROMETER FEATURES

The triple-axis MEMS accelerometer in ICM-40609-D includes a wide range of features:

- Digital-output X-, Y-, and Z-axis accelerometer with programmable full-scale range of $\pm 4g$, $\pm 8g$, $\pm 16g$ and $\pm 32g$
- Low Noise (LN) and Low Power (LP) power modes support
- User-programmable interrupts
- Wake-on-motion interrupt for low power operation of applications processor
- Self-test

2.3 ADDITIONAL FEATURES

ICM-40609-D includes the following additional features:

- 2KB FIFO buffer enables the applications processor to read the data in bursts
- User-programmable digital filters for gyroscope, accelerometer, and temperature sensor
- Wake on Motion: Detects motion when accelerometer data exceeds a programmable threshold
- 1 MHz I²C / 24 MHz SPI slave host interface
- Digital-output temperature sensor
- Smallest and thinnest LGA package for portable devices: 3 x 3 x 0.91 mm (24-pin LGA)
- 20,000g shock tolerant
- MEMS structure hermetically sealed and bonded at wafer level
- RoHS and Green compliant

3 ELECTRICAL CHARACTERISTICS

3.1 GYROSCOPE SPECIFICATIONS

Typical Operating Circuit of section 4.2, VDD = 1.8V, VDDIO = 1.8V, T_A=25°C, unless otherwise noted.

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
GYROSCOPE SENSITIVITY						
Full-Scale Range	GYRO_FS_SEL=0		±2000		°/s	2
	GYRO_FS_SEL =1		±1000		°/s	2
	GYRO_FS_SEL =2		±500		°/s	2
	GYRO_FS_SEL =3		±250		°/s	2
	GYRO_FS_SEL =4		±125		°/s	2
	GYRO_FS_SEL =5		±62.5		°/s	2
	GYRO_FS_SEL =6		±31.25		°/s	2
	GYRO_FS_SEL =7		±15.625		°/s	2
Gyroscope ADC Word Length			16		bits	2
Sensitivity Scale Factor	GYRO_FS_SEL=0		16.4		LSB/(°/s)	2
	GYRO_FS_SEL =1		32.8		LSB/(°/s)	2
	GYRO_FS_SEL =2		65.5		LSB/(°/s)	2
	GYRO_FS_SEL =3		131		LSB/(°/s)	2
	GYRO_FS_SEL =4		262		LSB/(°/s)	2
	GYRO_FS_SEL =5		524.3		LSB/(°/s)	2
	GYRO_FS_SEL =6		1048.6		LSB/(°/s)	2
	GYRO_FS_SEL =7		2097.2		LSB/(°/s)	2
Sensitivity Scale Factor Initial Tolerance	25°C		±0.5		%	1
Sensitivity Scale Factor Variation Over Temperature	-40°C to +85°C		±0.045		%/°C	3
Nonlinearity	Best fit straight line; 25°C		±0.1		%	3
Cross-Axis Sensitivity			±1		%	3
ZERO-RATE OUTPUT (ZRO)						
Initial ZRO Tolerance	Board-level, 25°C		±1		°/s	3
ZRO Variation vs. Temperature	-40°C to +85°C		±0.01		°/s/°C	3
OTHER PARAMETERS						
Rate Noise Spectral Density	@ 10 Hz		0.0045		°/s /√Hz	1
Total RMS Noise	Bandwidth = 100 Hz		0.045		°/s-rms	4
Gyroscope Mechanical Frequencies		25	27	29	KHz	1
Low Pass Filter Response	ODR < 1kHz	5		500	Hz	2
	ODR ≥ 1kHz	5		995	Hz	2
Gyroscope Start-Up Time	Time from gyro enable to gyro drive ready		30		ms	3
Output Data Rate		12.5		32000	Hz	2

Table 1. Gyroscope Specifications

Notes:

1. Tested in production.
2. Guaranteed by design.
3. Derived from validation or characterization of parts, not guaranteed in production.
4. Calculated from Rate Noise Spectral Density.

3.2 ACCELEROMETER SPECIFICATIONS

Typical Operating Circuit of section 4.2, VDD = 1.8V, VDDIO = 1.8V, T_A=25°C, unless otherwise noted.

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
ACCELEROMETER SENSITIVITY						
Full-Scale Range	ACCEL_FS_SEL =0		±32		g	2
	ACCEL_FS_SEL =1		±16		g	2
	ACCEL_FS_SEL =2		±8		g	2
	ACCEL_FS_SEL =3		±4		g	2
ADC Word Length	Output in two's complement format		16		bits	2
Sensitivity Scale Factor	ACCEL_FS_SEL =0		1,024		LSB/g	2
	ACCEL_FS_SEL =1		2,048		LSB/g	2
	ACCEL_FS_SEL =2		4,096		LSB/g	2
	ACCEL_FS_SEL =3		8,192		LSB/g	2
Sensitivity Scale Factor Initial Tolerance	Component-level		±0.5		%	1
Sensitivity Change vs. Temperature	-40°C to +85°C		±0.007		%/°C	3
Nonlinearity	Best Fit Straight Line, ±2g		±0.1		%	3
Cross-Axis Sensitivity			±1		%	3
ZERO-G OUTPUT						
Initial Tolerance	Board-level, all axes		±40		mg	3
Zero-G Level Change vs. Temperature	-40°C to +85°C		±0.15		mg/°C	3
OTHER PARAMETERS						
Power Spectral Density	@ 10 Hz		100		μg/√Hz	1
RMS Noise	Bandwidth = 100 Hz		1.00		mg-rms	4
Low-Pass Filter Response	ODR < 1kHz	5		500	Hz	2
	ODR ≥ 1kHz	5		995	Hz	2
Accelerometer Startup Time	From sleep mode to valid data		10		ms	3
Output Data Rate		1.5625		32000	Hz	2

Table 2. Accelerometer Specifications

Notes:

1. Tested in production.
2. Guaranteed by design.
3. Derived from validation or characterization of parts, not guaranteed in production.
4. Calculated from Power Spectral Density.

3.3 ELECTRICAL SPECIFICATIONS

3.3.1 D.C. Electrical Characteristics

Typical Operating Circuit of section 4.2, VDD = 1.8V, VDDIO = 1.8V, T_A=25°C, unless otherwise noted.

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
SUPPLY VOLTAGES						
VDD		1.71	1.8	3.6	V	1
VDDIO		1.71	1.8	3.6	V	1
SUPPLY CURRENTS						
Low-Noise Mode	6-Axis Gyroscope + Accelerometer		0.77		mA	2
	3-Axis Accelerometer		0.27		mA	2
	3-Axis Gyroscope		0.61		mA	2
Accelerometer Low -Power Mode (Gyroscope disabled)	200Hz ODR, 1x averaging		0.06		mA	2
Full-Chip Sleep Mode	At 25°C		11		μA	2
TEMPERATURE RANGE						
Specified Temperature Range	Performance parameters are not applicable beyond Specified Temperature Range	-40		+85	°C	1

Table 3. D.C. Electrical Characteristics

Notes:

1. Guaranteed by design.
2. Derived from validation or characterization of parts, not guaranteed in production.

3.3.2 A.C. Electrical Characteristics

Typical Operating Circuit of section 4.2, VDD = 1.8V, VDDIO = 1.8V, T_A=25°C, unless otherwise noted.

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
SUPPLIES						
Supply Ramp Time	Monotonic ramp. Ramp rate is 10% to 90% of the final value	0.01		3	ms	1
Power Supply Noise			10		mV peak-peak	1
TEMPERATURE SENSOR						
Operating Range	Ambient	-40		85	°C	1
25°C Output			0		LSB	3
ADC Resolution			16		bits	2
ODR	With Filter	25		8000	Hz	2
Room Temperature Offset	25°C	-5		5	°C	3
Stabilization Time				14000	μs	2
Sensitivity	Untrimmed		132.48		LSB/°C	1
Sensitivity for FIFO data			2.07		LSB/°C	1
POWER-ON RESET						
Start-up time for register read/write	From power-up			1	ms	1
I ² C ADDRESS						
I ² C ADDRESS	AP_AD0 = 0 AP_AD0 = 1		1101000 1101001			
DIGITAL INPUTS (FSYNC, SCLK, SDI, CS)						
V _{IH} , High Level Input Voltage		0.7*VDDIO			V	1
V _{IL} , Low Level Input Voltage				0.3*VDDIO	V	
C _i , Input Capacitance			< 10		pF	
DIGITAL OUTPUT (SDO, INT1, INT2)						
V _{OH} , High Level Output Voltage	R _{LOAD} =1 MΩ;	0.9*VDDIO			V	1
V _{OL1} , LOW-Level Output Voltage	R _{LOAD} =1 MΩ;			0.1*VDDIO	V	
V _{OLINT} , INT Low-Level Output Voltage	OPEN=1, 0.3 mA sink Current			0.1	V	
Output Leakage Current	OPEN=1		100		nA	
t _{INT} , INT Pulse Width	int_tpulse_duration= 0 , 1 (100us, 8us) ;	8		100	μs	
I ² C I/O (SCL, SDA)						
V _{IL} , LOW-Level Input Voltage		-0.5 V		0.3*VDDIO	V	1
V _{IH} , HIGH-Level Input Voltage		0.7*VDDIO		VDDIO + 0.5 V	V	
V _{hys} , Hysteresis			0.1*VDDIO		V	
V _{OL} , LOW-Level Output Voltage	3 mA sink current	0		0.4	V	
I _{OL} , LOW-Level Output Current	V _{OL} =0.4 V V _{OL} =0.6 V		3 6		mA mA	
Output Leakage Current			100		nA	
t _{of} , Output Fall Time from V _{IHmax} to V _{ILmax}	C _b bus capacitance in pf	20+0.1C _b		300	ns	
INTERNAL CLOCK SOURCE						
Clock Frequency Initial Tolerance	CLKSEL=`2b00 or gyro inactive; 25°C	-3		+3	%	1
	CLK_SEL=`2b01 and gyro active; 25°C	-1		+1	%	1
Frequency Variation over Temperature	CLK_SEL=`2b00 or gyro inactive; -40°C to +85°C			±3	%	1
	CLK_SEL=`2b01 and gyro active; -40°C to +85°C			±2	%	1

Table 4. A.C. Electrical Characteristics

Notes:

1. Expected results based on design, will be updated after characterization. Not guaranteed in production.
2. Guaranteed by design.
3. To be Production tested.

3.4 I²C TIMING CHARACTERIZATION

Typical Operating Circuit of section 4.2, VDD = 1.8V, VDDIO = 1.8V, T_A=25°C, unless otherwise noted.

PARAMETERS	CONDITIONS	MIN	TYPICAL	MAX	UNITS	NOTES
I²C TIMING						
I²C FAST-MODE PLUS						
f _{SCL} , SCL Clock Frequency				1	MHz	1
t _{HD,STA} , (Repeated) START Condition Hold Time		0.26			μs	1
t _{LOW} , SCL Low Period		0.5			μs	1
t _{HIGH} , SCL High Period		0.26			μs	1
t _{SU,STA} , Repeated START Condition Setup Time		0.26			μs	1
t _{HD,DAT} , SDA Data Hold Time		0			μs	1
t _{SU,DAT} , SDA Data Setup Time		50			ns	1
t _r , SDA and SCL Rise Time	C _b bus cap. from 10 to 400 pF			120	ns	1
t _f , SDA and SCL Fall Time	C _b bus cap. from 10 to 400 pF			120	ns	1
t _{SU,STO} , STOP Condition Setup Time		0.5			μs	1
t _{BUF} , Bus Free Time Between STOP and START Condition		0.5			μs	1
C _b , Capacitive Load for each Bus Line			< 400		pF	1
t _{VD,DAT} , Data Valid Time				0.45	μs	1
t _{VD,ACK} , Data Valid Acknowledge Time				0.45	μs	1

Table 5. I²C Timing Characteristics

Notes:

- Based on characterization of 5 parts over temperature and voltage as mounted on evaluation board or in sockets

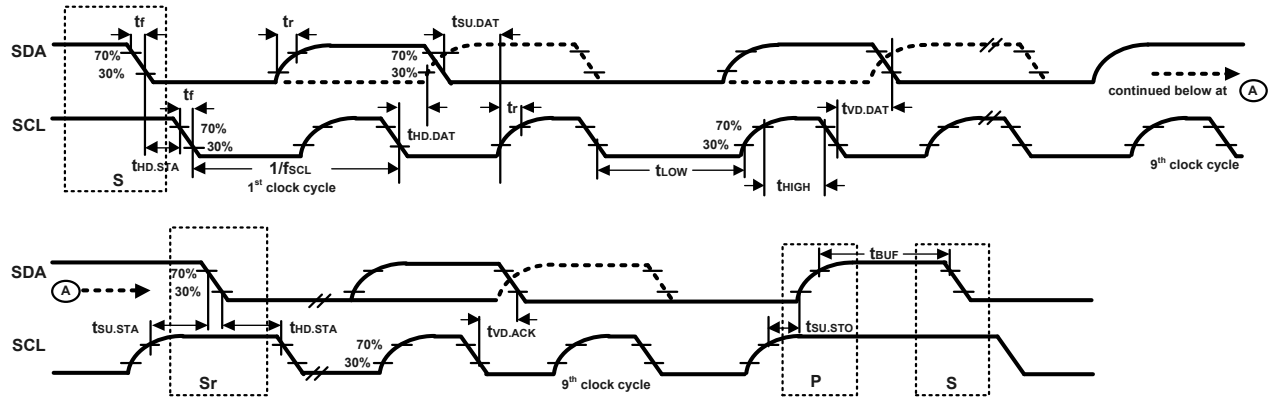


Figure 1. I²C Bus Timing Diagram

3.5 SPI TIMING CHARACTERIZATION – 4-WIRE SPI MODE

Typical Operating Circuit of section 4.2, VDD = 1.8V, VDDIO = 1.8V, T_A=25°C, unless otherwise noted.

PARAMETERS	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
SPI TIMING						
f _{SPC} , SCLK Clock Frequency	Default			24	MHz	1
t _{LOW} , SCLK Low Period		24.5			ns	1
t _{HIGH} , SCLK High Period		24.5			ns	1
t _{SU,CS} , CS Setup Time		39			ns	1
t _{HD,CS} , CS Hold Time		18			ns	1
t _{SU,SDI} , SDI Setup Time		13			ns	1
t _{HD,SDI} , SDI Hold Time		8			ns	1
t _{VD,SDO} , SDO Valid Time	C _{load} = 20 pF			21.5	ns	1
t _{HD,SDO} , SDO Hold Time	C _{load} = 20 pF	9.5			ns	1
t _{DIS,SDO} , SDO Output Disable Time				28	ns	1

Table 6. 4-Wire SPI Timing Characteristics (24-MHz Operation)

Notes:

1. Based on characterization of 5 parts over temperature and voltage as mounted on evaluation board or in sockets
2. Based on other parameter values

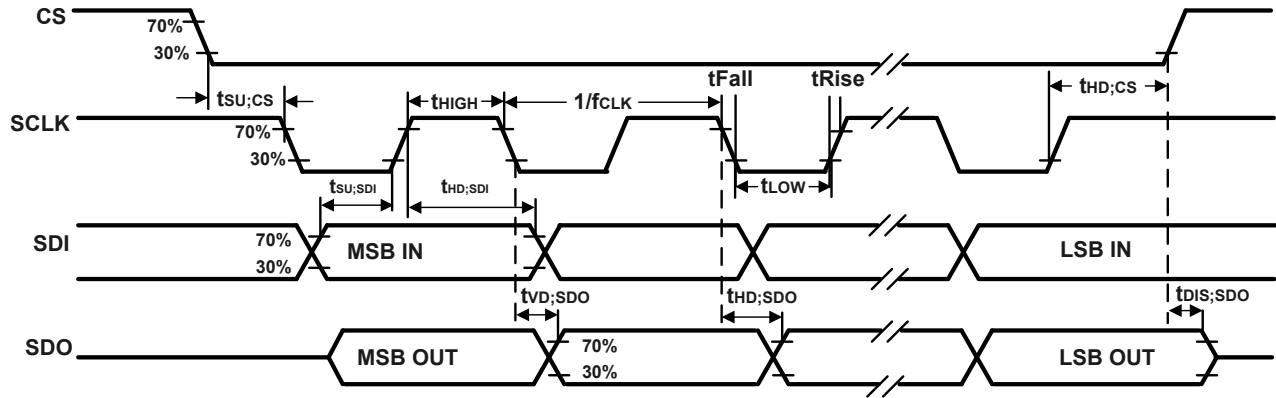


Figure 2. 4-Wire SPI Mode-3 Bus Timing Diagram

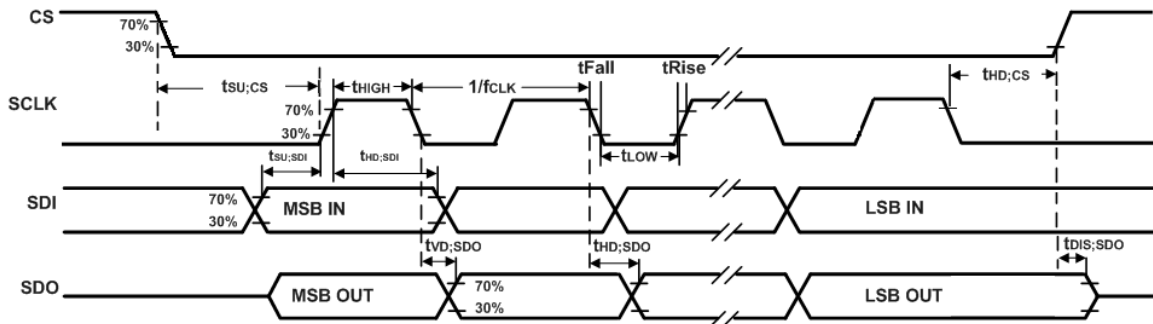


Figure 3. 4-Wire SPI Mode-0 Bus Timing Diagram

3.6 SPI TIMING CHARACTERIZATION – 3-WIRE SPI MODE

Typical Operating Circuit of section 4.2, VDD = 1.8V, VDDIO = 1.8V, T_A=25°C, unless otherwise noted.

PARAMETERS	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
SPI TIMING						
f _{SPC} , SCLK Clock Frequency	Default			24	MHz	1
t _{LOW} , SCLK Low Period		24.5			ns	1
t _{HIGH} , SCLK High Period		24.5			ns	1
t _{SU,CS} , CS Setup Time		39			ns	1
t _{HD,CS} , CS Hold Time		5			ns	1
t _{SU,SDIO} , SDIO Input Setup Time		13			ns	1
t _{HD,SDIO} , SDIO Input Hold Time		8			ns	1
t _{VD,SDIO} , SDIO Output Valid Time	C _{load} = 20 pF			18.5	ns	1
t _{HD,SDIO} , SDIO Output Hold Time	C _{load} = 20 pF	9.5			ns	1
t _{DIS,SDIO} , SDIO Output Disable Time				28	ns	1

Table 7. 3-Wire SPI Timing Characteristics (24-MHz Operation)

Notes:

1. Based on characterization of 5 parts over temperature and voltage as mounted on evaluation board or in sockets
2. Based on other parameter values

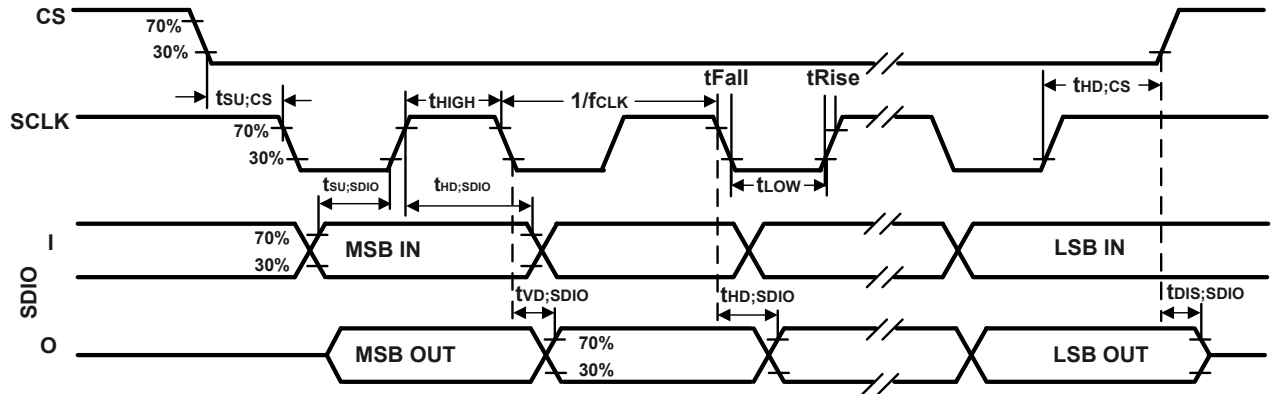


Figure 4. 3-Wire Mode-3 SPI Bus Timing Diagram

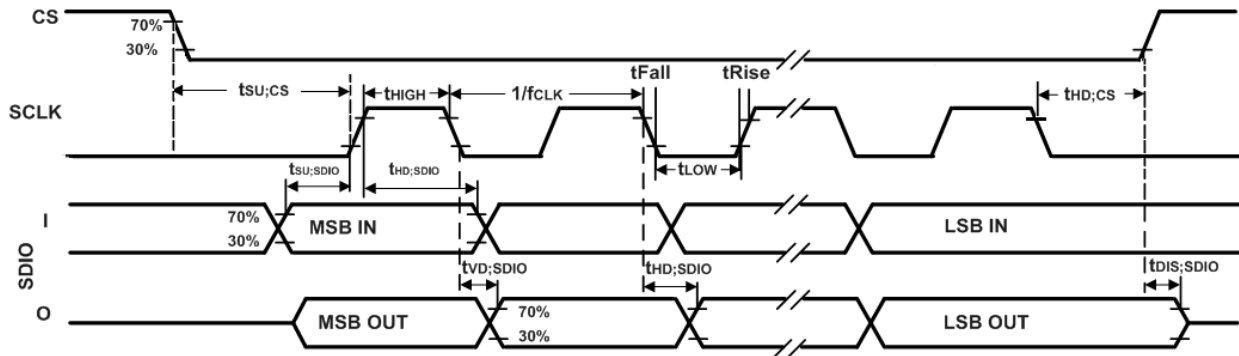


Figure 5. 3-Wire Mode-0 SPI Bus Timing Diagram

3.7 ABSOLUTE MAXIMUM RATINGS

Stress above those listed as “Absolute Maximum Ratings” may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these conditions is not implied. Exposure to the absolute maximum ratings conditions for extended periods may affect device reliability.

PARAMETER	RATING
Supply Voltage, VDD	-0.5V to +4V
Supply Voltage, VDDIO	-0.5V to +4V
Input Voltage Level (FSYNC, SCL, SDA)	-0.5V to VDDIO + 0.5V
Acceleration (Any Axis, unpowered)	20,000g for 0.2 ms
Operating Temperature Range	-40°C to +85°C
Storage Temperature Range	-40°C to +125°C
Electrostatic Discharge (ESD) Protection	2 kV (HBM); 200V (MM)
Latch-up	JEDEC Class II (2), 125°C ±100 mA

Table 8. Absolute Maximum Ratings

4 APPLICATIONS INFORMATION

4.1 PIN OUT DIAGRAM AND SIGNAL DESCRIPTION

PIN NUMBER	PIN NAME	PIN DESCRIPTION
8	VDDIO	Digital I/O supply voltage
9	AD0 / SDO	AD0: I ² C Slave Address LSB SDO: SPI serial data output (4-wire mode)
11	RESV	RESV: Connect to GND
12	INT1 / INT	INT1: Interrupt 1 INT: Interrupt if all interrupts mapped to pin 12
13	VDD	Power supply voltage
18	GND	Power supply ground
19	INT2 / FSYNC	INT2: Interrupt 2 FSYNC: Frame sync input; Connect to GND if FSYNC selected but FSYNC signal not input
20	RESV	Reserved, connect to GND
22	nCS	Chip select (SPI mode only)
23	SCL / SCLK	SCL: I ² C serial clock SCLK: SPI serial clock
24	SDA / SDI / SDIO	SDA: I ² C serial data SDI: SPI serial data input (4-wire mode) SDIO: SPI serial data I/O (3-wire mode)
1 – 7, 10, 14 – 17, 21	NC	No Connect pins. Do not connect.

Table 9. Signal Descriptions

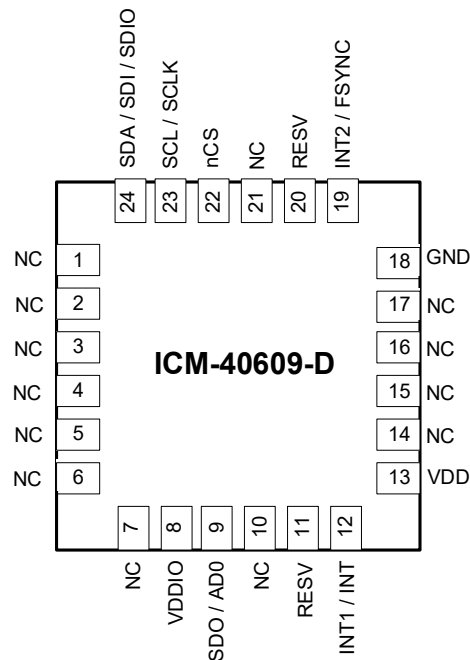


Figure 6. Pin Out Diagram for ICM-40609-D 3x3.0x0.91 mm LGA

4.2 TYPICAL OPERATING CIRCUIT

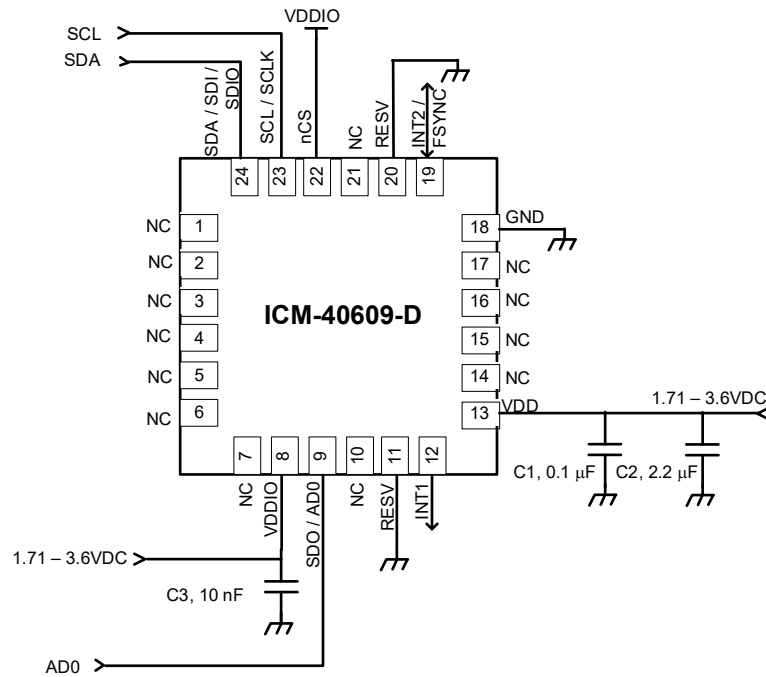


Figure 7. ICM-40609-D Application Schematic (I²C Interface to Host)

Note: I²C lines are open drain and pull-up resistors (e.g. 10 kΩ) are required.

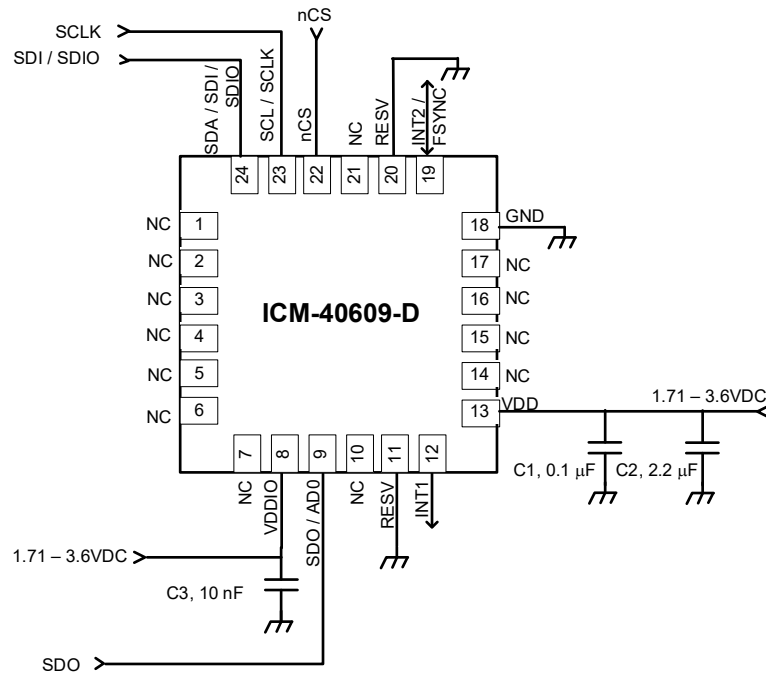


Figure 8. ICM-40609-D Application Schematic (SPI Interface to Host)

4.3 BILL OF MATERIALS FOR EXTERNAL COMPONENTS

COMPONENT	LABEL	SPECIFICATION	QUANTITY
VDD Bypass Capacitors	C1	X7R, 0.1 μ F $\pm$ 10%	1
	C2	X7R, 2.2 μ F $\pm$ 10%	1
VDDIO Bypass Capacitor	C3	X7R, 10nF $\pm$ 10%	1

Table 10. Bill of Materials

4.4 SYSTEM BLOCK DIAGRAM

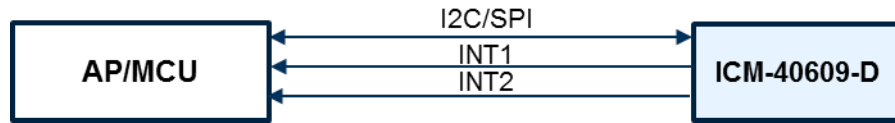


Figure 9. ICM-40609-D System Block Diagram

Note: The block diagram in Figure 9 is an example. Please refer to the pin-out (section 4.1) for other configuration options.

4.5 OVERVIEW

The ICM-40609-D is comprised of the following key blocks and functions:

- Three-axis MEMS rate gyroscope sensor with 16-bit ADCs and signal conditioning
- Three-axis MEMS accelerometer sensor with 16-bit ADCs and signal conditioning
- I²C and SPI serial communications interfaces to Host
- Self-Test
- Clocking
- Sensor Data Registers
- FIFO
- Interrupts
- Digital-Output Temperature Sensor
- Bias and LDOs
- Charge Pump
- Standard Power Modes

4.6 THREE-AXIS MEMS GYROSCOPE WITH 16-BIT ADCS AND SIGNAL CONDITIONING

The ICM-40609-D includes a vibratory MEMS rate gyroscope, which detects rotation about the X-, Y-, and Z- Axes. When the gyroscope is rotated about any of the sense axes, the Coriolis Effect causes a vibration that is detected by a capacitive pickoff. The resulting signal is amplified, demodulated, and filtered to produce a voltage that is proportional to the angular rate. This voltage is digitized using on-chip Analog-to-Digital Converters (ADCs) to sample each axis. The full-scale range of the gyro sensors may be digitally programmed to ± 15.625 , ± 31.25 , ± 62.5 , ± 125 , ± 250 , ± 500 , ± 1000 , and ± 2000 degrees per second (dps).

4.7 THREE-AXIS MEMS ACCELEROMETER WITH 16-BIT ADCS AND SIGNAL CONDITIONING

The ICM-40609-D includes a 3-Axis MEMS accelerometer. Acceleration along a particular axis induces displacement of a proof mass in the MEMS structure, and capacitive sensors detect the displacement. The ICM-40609-D architecture reduces the accelerometers' susceptibility to fabrication variations as well as to thermal drift. When the device is placed on a flat surface, it will measure 0g on the X- and Y-axes and +1g on the Z-axis. The accelerometers' scale factor is calibrated at the factory and is nominally independent of supply voltage. The full-scale range of the digital output can be adjusted to $\pm 4g$, $\pm 8g$, $\pm 16g$ and $\pm 32g$.

4.8 I²C AND SPI HOST INTERFACE

The ICM-40609-D communicates to the application processor using an I²C, or SPI serial interface. The ICM-40609-D always acts as a slave when communicating to the application processor.

4.9 SELF-TEST

Self-test allows for the testing of the mechanical and electrical portions of the sensors. The self-test for each measurement axis can be activated by means of the gyroscope and accelerometer self-test registers.

When the self-test is activated, the electronics cause the sensors to be actuated and produce an output signal. The output signal is used to observe the self-test response.

The self-test response is defined as follows:

$$\text{SELF-TEST RESPONSE} = \text{SENSOR OUTPUT WITH SELF-TEST ENABLED} - \text{SENSOR OUTPUT WITH SELF-TEST DISABLED}$$

When the value of the self-test response is within the specified min/max limits of the product specification, the part has passed self-test. When the self-test response exceeds the min/max values, the part is deemed to have failed self-test.

4.10 CLOCKING

The ICM-40609-D has a flexible clocking scheme, allowing the following internal clock sources to be used for the internal synchronous circuitry. This synchronous circuitry includes the signal conditioning and ADCs, and various control circuits and registers.

- a) An internal relaxation oscillator
- b) Auto-select between internal relaxation oscillator and gyroscope MEMS oscillator to use the best available source

The only setting supporting specified performance in all modes is option b). It is recommended that option b) be used when using internal clock source.

4.11 SENSOR DATA REGISTERS

The sensor data registers contain the latest gyroscope, accelerometer, and temperature measurement data. They are read-only registers and are accessed via the serial interface. Data from these registers may be read anytime.

4.12 INTERRUPTS

Interrupt functionality is configured via the Interrupt Configuration register. Items that are configurable include the interrupt pins configuration, the interrupt latching and clearing method, and triggers for the interrupt. Items that can trigger an interrupt are (1) Clock generator locked to new reference oscillator (used when switching clock sources); (2) new data is available to be read (from the FIFO and Data registers); (3) accelerometer event interrupts; (4) FIFO watermark; (5) FIFO overflow. The interrupt status can be read from the Interrupt Status register.

4.13 DIGITAL-OUTPUT TEMPERATURE SENSOR

An on-chip temperature sensor and ADC are used to measure the ICM-40609-D die temperature. The readings from the ADC can be read from the FIFO or the Sensor Data registers.

Temperature data value from the sensor data registers can be converted to degrees centigrade by using the following formula:

$$\text{Temperature in Degrees Centigrade} = (\text{TEMP_DATA} / 132.48) + 25$$

Temperature data stored in FIFO is an 8-bit quantity, FIFO_TEMP_DATA. It can be converted to degrees centigrade by using the following formula:

$$\text{Temperature in Degrees Centigrade} = (\text{FIFO_TEMP_DATA} / 2.07) + 25$$

4.14 BIAS AND LDOS

The bias and LDO section generates the internal supply and the reference voltages and currents required by the ICM-40609-D.

4.15 CHARGE PUMP

An on-chip charge pump generates the high voltage required for the MEMS oscillator.

4.16 STANDARD POWER MODES

Table 11 lists the user-accessible power modes for ICM-40609-D.

MODE	NAME	GYRO	ACCEL
1	Sleep Mode	Off	Off
2	Standby Mode	Drive On	Off
3	Accelerometer Low-Power Mode	Off	Duty-Cycled
4	Accelerometer Low-Noise Mode	Off	On
5	Gyroscope Low-Noise Mode	On	Off
6	6-Axis Low-Noise Mode	On	On

Table 11. Standard Power Modes for ICM-40609-D

5 SIGNAL PATH

Figure 10 shows a block diagram of the signal path for ICM-40609-D.

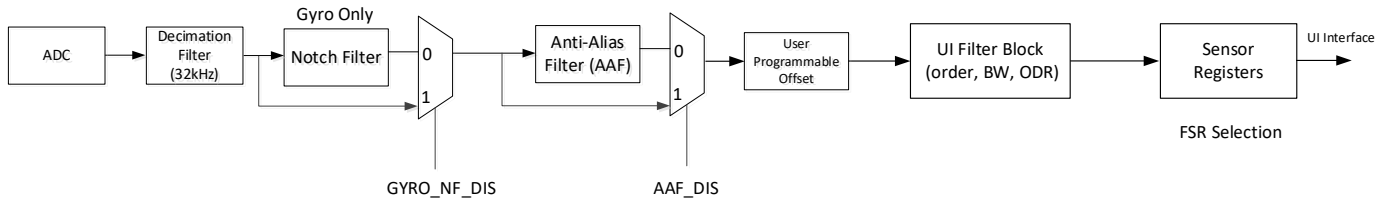


Figure 10. ICM-40609-D Signal Path

The signal path starts with independent 16-bit ADCs for each axis of gyroscope and accelerometer. The ADC output goes through a Decimation Filter that has fixed bandwidth of 32 kHz for ICM-40609-D. Other components of the signal path are described below in further detail.

5.1 NOTCH FILTER

The Notch Filter is supported only for the gyroscope signal path. The following steps can be used to program the notch filter. Note that the notch filter is specific to each axis in the gyroscope, so the X, Y, and Z axis can be programmed independently.

5.1.1 Frequency of Notch Filter (each axis)

To operate the Notch filter, two parameters NF_COSWZ, and NF_COSWZ_SEL must be programmed for each gyroscope axis.

Parameters NF_COSWZ are defined for each axis of the gyroscope as GYRO_X_NF_COSWZ (register bank 1, register 0x0Fh & register 0x12h), GYRO_Y_NF_COSWZ (register bank 1, register 0x10h & register 0x12h), GYRO_Z_NF_COSWZ (register bank 1, register 0x11h & register 0x12h). Note that the parameters have 9-bit values across two different registers.

Parameters NF_COSWZ_SEL are defined for each axis of the gyroscope as GYRO_X_NF_COSWZ_SEL (register bank 1, register 0x12h, bit 3), GYRO_Y_NF_COSWZ_SEL (register bank 1, register 0x12h, bit 4), GYRO_Z_NF_COSWZ_SEL (register bank 1, register 0x12h, bit 5).

Each value must be calculated using the steps described below and programmed into the corresponding register locations mentioned above.

$f_{desired}$ is the desired frequency of the Notch Filter in kHz. The lower bound for $f_{desired}$ is 1 kHz, and the upper bound is 3 kHz. Operating the notch filter outside this range is not supported.

Step1: $COSWZ = \cos(2 \cdot \pi \cdot f_{desired} / 32)$

Step2:

```

If abs(COSWZ) ≤ 0.875
    NF_COSWZ = round[COSWZ*256]
    NF_COSWZ_SEL = 0
else
    NF_COSWZ_SEL = 1
    if COSWZ > 0.875
        NF_COSWZ = round [8*(1-COSWZ)*256]
    else if COSWZ < -0.875
        NF_COSWZ = round [-8*(1+COSWZ)*256]
    end
End
    
```


5.1.2 Bandwidth of Notch Filter (common to all axes)

The notch filter allows the user to control the width of the notch from eight possible values using a 3-bit parameter GYRO_NF_BW_SEL in register bank 1, register 0x13h, bits 6:4. This parameter is common to all three axes.

GYRO_NF_BW_SEL	Notch Filter Bandwidth (Hz)
0	1449
1	680
2	329
3	162
4	80
5	40
6	20
7	10

The notch filter can be selected or bypassed by using the parameter GYRO_NF_DIS in register bank 1, register 0x0Bh, bit 0 as shown below.

GYRO_NF_DIS	Function
0	Enable notch filter
1	Disable notch filter

5.2 ANTI-ALIAS FILTER

Anti-alias filters for gyroscope and accelerometer can be independently programmed to have bandwidths ranging from 10 Hz to 995 Hz. To program the anti-alias filter for a required bandwidth, use the table below to map the bandwidth to register values as shown:

- Register bank 2, register 0x03h, bits 6:1, ACCEL_AAF_DELT: Code from 1 to 63 that allows programming the bandwidth for accelerometer anti-alias filter
- Register bank 2, register 0x04h, bits 7:0 and Bank 2, register 0x05h, bits 3:0, ACCEL_AAF_DELTSQR: Square of the delt value for accelerometer
- Register bank 2, register 0x05h, bits 7:4, ACCEL_AAF_BITSHIFT: Bitshift value for accelerometer used in hardware implementation
- Register bank 1, register 0x0Ch, bits 5:0, GYRO_AAF_DELT: Code from 1 to 63 that allows programming the bandwidth for gyroscope anti-alias filter
- Register bank 1, register 0x0Dh, bits 7:0 and Bank 1, register 0x0Eh, bits 3:0, GYRO_AAF_DELTSQR: Square of the delt value for gyroscope
- Register bank 1, register 0x0Eh, bits 7:4, GYRO_AAF_BITSHIFT: Bitshift value for gyroscope used in hardware implementation

3dB Bandwidth (Hz)	ACCEL_AAF_DELT; GYRO_AAF_DELT	ACCEL_AAF_DELTSQR; GYRO_AAF_DELTSQR	ACCEL_AAF_BITSHIFT; GYRO_AAF_BITSHIFT
42	1	1	15
84	2	4	13
126	3	9	12
170	4	16	11
213	5	25	10
258	6	36	10
303	7	49	9
348	8	64	9
394	9	81	9
441	10	100	8
488	11	122	8
536	12	144	8
585	13	170	8
634	14	196	7
684	15	224	7
734	16	256	7
785	17	288	7
837	18	324	7
890	19	360	6
943	20	400	6
997	21	440	6
1051	22	488	6
1107	23	528	6
1163	24	576	6
1220	25	624	6
1277	26	680	6
1336	27	736	5
1395	28	784	5
1454	29	848	5
1515	30	896	5
1577	31	960	5
1639	32	1024	5
1702	33	1088	5
1766	34	1152	5
1830	35	1232	5
1896	36	1296	5
1962	37	1376	4
2029	38	1440	4
2097	39	1536	4
2166	40	1600	4

2235	41	1696	4
2306	42	1760	4
2377	43	1856	4
2449	44	1952	4
2522	45	2016	4
2596	46	2112	4
2671	47	2208	4
2746	48	2304	4
2823	49	2400	4
2900	50	2496	4
2978	51	2592	4
3057	52	2720	4
3137	53	2816	3
3217	54	2944	3
3299	55	3008	3
3381	56	3136	3
3464	57	3264	3
3548	58	3392	3
3633	59	3456	3
3718	60	3584	3
3805	61	3712	3
3892	62	3840	3
3979	63	3968	3

The anti-alias filter can be selected or bypassed for the gyroscope by using the parameter GYRO_AAF_DIS in register bank 1, register 0x0Bh, bit 1 as shown below.

GYRO_AAF_DIS	Function
0	Enable gyroscope anti-aliasing filter
1	Disable gyroscope anti-aliasing filter

The anti-alias filter can be selected or bypassed for the accelerometer by using the parameter ACCEL_AAF_DIS in register bank 2, register 0x03h, bit 0 as shown below.

ACCEL_AAF_DIS	Function
0	Enable accelerometer anti-aliasing filter
1	Disable accelerometer anti-aliasing filter

5.3 USER PROGRAMMABLE OFFSET

Gyroscope and accelerometer offsets can be programmed by the user by using registers OFFSET_USER0, through OFFSET_USER8, in bank 0, registers 0x77h through 0x7Fh (bank 4) as shown below.

REGISTER ADDRESS	REGISTER NAME	BITS	FUNCTION
0x77h	OFFSET_USER0	7:0	Lower bits of X-gyro offset programmed by user. Max value is ± 64 dps, resolution is 1/32 dps.
0x78h	OFFSET_USER1	3:0	Upper bits of X-gyro offset programmed by user. Max value is ± 64 dps, resolution is 1/32 dps.
		7:4	Upper bits of Y-gyro offset programmed by user. Max value is ± 64 dps, resolution is 1/32 dps.
0x79h	OFFSET_USER2	7:0	Lower bits of Y-gyro offset programmed by user. Max value is ± 64 dps, resolution is 1/32 dps.
0x7Ah	OFFSET_USER3	7:0	Lower bits of Z-gyro offset programmed by user. Max value is ± 64 dps, resolution is 1/32 dps.
0x7Bh	OFFSET_USER4	3:0	Upper bits of Z-gyro offset programmed by user. Max value is ± 64 dps, resolution is 1/32 dps.
		7:4	Upper bits of X-accel offset programmed by user. Max value is ± 1 g, resolution is 0.5 g.
0x7Ch	OFFSET_USER5	7:0	Lower bits of X-accel offset programmed by user. Max value is ± 1 g, resolution is 0.5 g.
0x7Dh	OFFSET_USER6	7:0	Lower bits of Y-accel offset programmed by user. Max value is ± 1 g, resolution is 0.5 g.
0x7Eh	OFFSET_USER7	3:0	Upper bits of Y-accel offset programmed by user. Max value is ± 1 g, resolution is 0.5 g.
		7:4	Upper bits of Z-accel offset programmed by user. Max value is ± 1 g, resolution is 0.5 g.
0x7Fh	OFFSET_USER8	7:0	Lower bits of Z-accel offset programmed by user. Max value is ± 1 g, resolution is 0.5 g.

5.4 UI FILTER BLOCK

The UI filter block can be programmed to select filter order and bandwidth independently for gyroscope and accelerometer.

Gyroscope filter order can be selected by programming the parameter GYRO_UI_FILT_ORD in register bank 0, register 0x51h, bits 3:2, as shown below.

GYRO_UI_FILT_ORD	Filter Order
00	1 st order
01	2 nd order
10	3 rd order
11	Reserved

Accelerometer filter order can be selected by programming the parameter ACCEL_UI_FILT_ORD in register bank 0, register 0x53h, bits 4:3, as shown below.

ACCEL_UI_FILT_ORD	Filter Order
00	1 st order
01	2 nd order
10	3 rd order
11	Reserved

Gyroscope and accelerometer filter 3dB bandwidth can be selected by programming the parameter GYRO_UI_FILT_BW in register bank 0, register 0x52h, bits 3:0, and the parameter ACCEL_UI_FILT_BW in register bank 0, register 0x52h, bits 7:4, as shown below. The values shown in bold correspond to low noise and the values shown in italics correspond to low latency. User can select the appropriate setting based on the application requirements for power and latency. Corresponding Noise Bandwidth (NBW) and Group Delay values are also shown.

5.4.1 1st Order Filter 3dB Bandwidth, Noise Bandwidth (NBW), Group Delay

3dB Bandwidth (Hz) for GYRO/ACCEL_UI_FILT_ORD=0 (1st order filter)											
		GYRO/ACCEL_UI_FILT_BW									
GYRO/ACCEL_ODR	ODR(Hz)	0	1	2	3	4	5	6	7	14	15
1	32000	8400.0									
2	16000	4194.1									
3	8000	2096.3									
4	4000	1048.1									
5	2000	524.0									
6	1000	498.3	227.2	188.9	111.0	92.4	59.6	48.8	23.9	262.0	2096.3
15	500	249.1	113.6	94.4	55.5	46.2	29.8	24.4	11.9	131.0	1048.1
7	200	99.6	90.9	75.5	44.4	37.0	23.8	19.5	9.6	104.8	419.2
8	100	49.8	90.9	75.5	44.4	37.0	23.8	19.5	9.6	104.8	209.6
9	50	24.9	90.9	75.5	44.4	37.0	23.8	19.5	9.6	104.8	104.8
10	25	12.5	90.9	75.5	44.4	37.0	23.8	19.5	9.6	104.8	52.4
11	12.5	12.5	90.9	75.5	44.4	37.0	23.8	19.5	9.6	104.8	52.4

NBW Bandwidth (Hz) for GYRO/ACCEL_UI_FILT_ORD=0 (1st order filter)											
		GYRO/ACCEL_UI_FILT_BW									
GYRO/ACCEL_ODR	ODR(Hz)	0	1	2	3	4	5	6	7	14	15
1	32000	8831.7									
2	16000	4410.6									
3	8000	2204.6									
4	4000	1102.2									
5	2000	551.1									
6	1000	551.1	230.8	196.3	126.5	108.9	75.8	64.1	34.1	275.6	2204.6
15	500	280.5	115.4	98.2	63.3	54.5	37.9	32.1	17.1	137.8	1102.2
7	200	112.2	92.4	78.5	50.6	43.6	30.3	25.7	13.7	110.3	440.9
8	100	56.1	92.4	78.5	50.6	43.6	30.3	25.7	13.7	110.3	220.5
9	50	28.1	92.4	78.5	50.6	43.6	30.3	25.7	13.7	110.3	110.3
10	25	14.1	92.4	78.5	50.6	43.6	30.3	25.7	13.7	110.3	55.2
11	12.5	14.1	92.4	78.5	50.6	43.6	30.3	25.7	13.7	110.3	55.2

Group Delay @DC (ms) for GYRO/ACCEL_UI_FILT_ORD=0 (1st order filter)											
		GYRO/ACCEL_UI_FILT_BW									
GYRO/ACCEL_ODR	ODR(Hz)	0	1	2	3	4	5	6	7	14	15
1	32000	0.1									

2	16000	0.1									
3	8000	0.2									
4	4000	0.4									
5	2000	0.8									
6	1000	0.6	1.8	2.0	2.8	3.1	4.1	4.7	8.1	1.5	0.2
15	500	1.1	3.6	4.0	5.5	6.1	8.1	9.3	16.2	3.0	0.4
7	200	2.7	4.4	5.0	6.8	7.6	10.2	11.7	20.3	3.8	1.0
8	100	5.3	4.4	5.0	6.8	7.6	10.2	11.7	20.3	3.8	1.9
9	50	10.5	4.4	5.0	6.8	7.6	10.2	11.7	20.3	3.8	3.8
10	25	21.0	4.4	5.0	6.8	7.6	10.2	11.7	20.3	3.8	7.5
11	12.5	21.0	4.4	5.0	6.8	7.6	10.2	11.7	20.3	3.8	7.5

5.4.2 2nd Order Filter 3dB Bandwidth, Noise Bandwidth (NBW), Group Delay

		3dB Bandwidth (Hz) for GYRO/ACCEL_UI_FILT_ORD=1 (2nd order filter)									
		GYRO/ACCEL_UI_FILT_BW									
GYRO/ACCEL_ODR	ODR(Hz)	0	1	2	3	4	5	6	7	14	15
1	32000	8400.0									
2	16000	4194.1									
3	8000	2096.3									
4	4000	1048.1									
5	2000	524.0									
6	1000	493.3	230.7	191.6	117.5	97.1	59.6	48.0	21.3	262.0	2096.3
15	500	246.7	115.3	95.8	58.8	48.5	29.8	24.0	10.6	131.0	1048.1
7	200	98.7	92.3	76.6	47.0	38.8	23.8	19.2	8.5	104.8	419.2
8	100	49.3	92.3	76.6	47.0	38.8	23.8	19.2	8.5	104.8	209.6
9	50	24.7	92.3	76.6	47.0	38.8	23.8	19.2	8.5	104.8	104.8
10	25	12.3	92.3	76.6	47.0	38.8	23.8	19.2	8.5	104.8	52.4
11	12.5	12.3	92.3	76.6	47.0	38.8	23.8	19.2	8.5	104.8	52.4

NBW Bandwidth (Hz) for GYRO/ACCEL_UI_FILT_ORD=1 (2nd order filter)											
		GYRO/ACCEL_UI_FILT_BW									
GYRO/ACCEL_ODR	ODR(Hz)	0	1	2	3	4	5	6	7	14	15
1	32000	8831.7									
2	16000	4410.6									
3	8000	2204.6									
4	4000	1102.2									
5	2000	551.1									
6	1000	551.1	223.7	189.9	122.7	102.8	64.7	52.5	23.7	275.6	2204.6
15	500	259.6	111.9	95.0	61.4	51.4	32.4	26.3	11.9	137.8	1102.2
7	200	103.9	89.5	76.0	49.1	41.2	25.9	21.0	9.5	110.3	440.9
8	100	52.0	89.5	76.0	49.1	41.2	25.9	21.0	9.5	110.3	220.5
9	50	26.0	89.5	76.0	49.1	41.2	25.9	21.0	9.5	110.3	110.3
10	25	13.0	89.5	76.0	49.1	41.2	25.9	21.0	9.5	110.3	55.2
11	12.5	13.0	89.5	76.0	49.1	41.2	25.9	21.0	9.5	110.3	55.2

Group Delay @DC (ms) for GYRO/ACCEL_UI_FILT_ORD=1 (2nd order filter)											
		GYRO/ACCEL_UI_FILT_BW									
GYRO/ACCEL_ODR	ODR(Hz)	0	1	2	3	4	5	6	7	14	15
1	32000	0.1									
2	16000	0.1									
3	8000	0.2									
4	4000	0.4									
5	2000	0.8									
6	1000	0.7	2.1	2.4	3.2	3.7	5.2	6.1	12.0	1.5	0.2
15	500	1.3	4.1	4.7	6.4	7.3	10.4	12.2	24.0	3.0	0.4
7	200	3.3	5.1	5.8	8.0	9.1	12.9	15.3	30.0	3.8	1.0
8	100	6.5	5.1	5.8	8.0	9.1	12.9	15.3	30.0	3.8	1.9
9	50	12.9	5.1	5.8	8.0	9.1	12.9	15.3	30.0	3.8	3.8
10	25	25.7	5.1	5.8	8.0	9.1	12.9	15.3	30.0	3.8	7.5
11	12.5	25.7	5.1	5.8	8.0	9.1	12.9	15.3	30.0	3.8	7.5

5.4.3 3rd Order Filter 3dB Bandwidth, Noise Bandwidth (NBW), Group Delay

3dB Bandwidth (Hz) for GYRO/ACCEL_UI_FILT_ORD=2 (3rd order filter)											
		GYRO/ACCEL_UI_FILT_BW									
GYRO/ACCEL_ODR	ODR(Hz)	0	1	2	3	4	5	6	7	14	15
1	32000	8400.0									
2	16000	4194.1									
3	8000	2096.3									
4	4000	1048.1									
5	2000	524.0									
6	1000	492.9	234.7	195.8	118.9	97.9	60.8	46.8	25.2	262.0	2096.3
15	500	246.4	117.4	97.9	59.5	48.9	30.4	23.4	12.6	131.0	1048.1
7	200	98.6	93.9	78.3	47.6	39.2	24.3	18.7	10.1	104.8	419.2
8	100	49.3	93.9	78.3	47.6	39.2	24.3	18.7	10.1	104.8	209.6
9	50	24.6	93.9	78.3	47.6	39.2	24.3	18.7	10.1	104.8	104.8
10	25	12.3	93.9	78.3	47.6	39.2	24.3	18.7	10.1	104.8	52.4
11	12.5	12.3	93.9	78.3	47.6	39.2	24.3	18.7	10.1	104.8	52.4

NBW Bandwidth (Hz) for GYRO/ACCEL_UI_FILT_ORD=0 (1st order filter)											
		GYRO/ACCEL_UI_FILT_BW									
GYRO/ACCEL_ODR	ODR(Hz)	0	1	2	3	4	5	6	7	14	15
1	32000	8831.7									
2	16000	4410.6									
3	8000	2204.6									
4	4000	1102.2									
5	2000	551.1									
6	1000	551.1	221.3	188.5	120.1	100.0	62.9	48.6	26.4	275.6	2204.6
15	500	252.0	110.7	94.3	60.1	50.0	31.5	24.3	13.2	137.8	1102.2
7	200	100.8	88.6	75.4	48.1	40.0	25.2	19.5	10.6	110.3	440.9
8	100	50.4	88.6	75.4	48.1	40.0	25.2	19.5	10.6	110.3	220.5
9	50	25.2	88.6	75.4	48.1	40.0	25.2	19.5	10.6	110.3	110.3
10	25	12.6	88.6	75.4	48.1	40.0	25.2	19.5	10.6	110.3	55.2
11	12.5	12.6	88.6	75.4	48.1	40.0	25.2	19.5	10.6	110.3	55.2

Group Delay @DC (ms) for GYRO/ACCEL_UI_FILT_ORD=2 (3rd order filter)											
		GYRO/ACCEL_UI_FILT_BW									
GYRO/ACCEL_ODR	ODR(Hz)	0	1	2	3	4	5	6	7	14	15
1	32000	0.1									
2	16000	0.1									
3	8000	0.2									
4	4000	0.4									
5	2000	0.8									
6	1000	0.8	2.3	2.7	4.0	4.6	6.6	8.2	14.1	1.5	0.2
15	500	1.6	4.6	5.4	7.9	9.2	13.2	16.3	28.1	3.0	0.4
7	200	4.0	5.8	6.8	9.8	11.4	16.5	20.4	35.2	3.8	1.0
8	100	8.0	5.8	6.8	9.8	11.4	16.5	20.4	35.2	3.8	1.9
9	50	15.9	5.8	6.8	9.8	11.4	16.5	20.4	35.2	3.8	3.8
10	25	31.8	5.8	6.8	9.8	11.4	16.5	20.4	35.2	3.8	7.5
11	12.5	31.8	5.8	6.8	9.8	11.4	16.5	20.4	35.2	3.8	7.5

5.5 ODR AND FSR SELECTION

Gyroscope ODR can be selected by programming the parameter GYRO_ODR in register bank 0, register 0x4Fh, bits 3:0 as shown below.

GYRO_ODR	Gyroscope ODR Value
0000	Reserved
0001	32 kHz
0010	16 kHz
0011	8 kHz
0100	4 kHz
0101	2 kHz
0110	1 kHz (default)
0111	200 Hz
1000	100 Hz
1001	50 Hz
1010	25 Hz
1011	12.5 Hz
1100	Reserved
1101	Reserved
1110	Reserved
1111	500 Hz

Gyroscope FSR can be selected by programming the parameter GYRO_FS_SEL in register bank 0, register 0x4Fh, bits 7:5 as shown below.

GYRO_FS_SEL	Gyroscope FSR Value
000	±2000 dps
001	±1000 dps
010	±500 dps
011	±250 dps
100	±125 dps
101	±62.5 dps
110	±31.25 dps
111	±15.625 dps

Accelerometer ODR can be selected by programming the parameter ACCEL_ODR in register bank 0, register 0x50h, bits 3:0 as shown below.

ACCEL_ODR	Accelerometer ODR Value
0000	Reserved
0001	32 kHz
0010	16 kHz
0011	8 kHz
0100	4 kHz
0101	2 kHz
0110	1 kHz (default)
0111	200 Hz
1000	100 Hz
1001	50 Hz
1010	25 Hz
1011	12.5 Hz
1100	6.25 Hz
1101	3.125 Hz
1110	1.5625 Hz
1111	500 Hz

Accelerometer FSR can be selected by programming the parameter ACCEL_FS_SEL in register bank 0, register 0x50h, bits 7:5 as shown below.

ACCEL_FS_SEL	Accelerometer FSR Value
000	±32g
001	±16g
010	±8g
011	±4g
100	Reserved
101	Reserved
110	Reserved
111	Reserved

6 FIFO

The ICM-40609-D contains a 2 KB FIFO register that is accessible via the serial interface. The FIFO configuration register determines which data is written into the FIFO. Possible choices include gyroscope data, accelerometer data, temperature readings, and FSYNC input. A FIFO counter keeps track of how many bytes of valid data are contained in the FIFO.

6.1 PACKET STRUCTURE

The following figure shows the FIFO packet structures supported in ICM-40609-D.

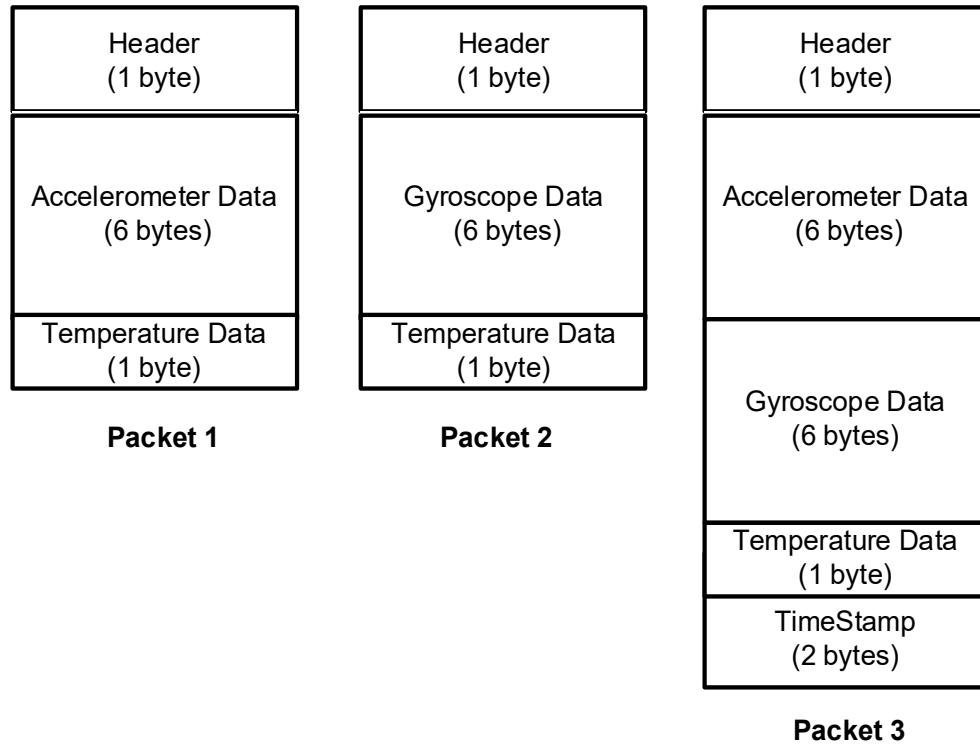


Figure 11. FIFO Packet Structure

The rest of this sub-section describes how individual data is packaged in the different FIFO packet structures.

Packet 1: Individual data is packaged in Packet 1 as shown below.

Byte	Content
0x00	FIFO Header
0x01	Accel X [15:8]
0x02	Accel X [7:0]
0x03	Accel Y [15:8]
0x04	Accel Y [7:0]
0x05	Accel Z [15:8]
0x06	Accel Z [7:0]
0x07	Temperature[7:0]

Packet 2: Individual data is packaged in Packet 2 as shown below.

Byte	Content
0x00	FIFO Header
0x01	Gyro X [15:8]
0x02	Gyro X [7:0]
0x03	Gyro Y [15:8]
0x04	Gyro Y [7:0]
0x05	Gyro Z [15:8]
0x06	Gyro Z [7:0]
0x07	Temperature[7:0]

Packet 3: Individual data is packaged in Packet 3 as shown below.

Byte	Content
0x00	FIFO Header
0x01	Accel X [15:8]
0x02	Accel X [7:0]
0x03	Accel Y [15:8]
0x04	Accel Y [7:0]
0x05	Accel Z [15:8]
0x06	Accel Z [7:0]
0x07	Gyro X [15:8]
0x08	Gyro X [7:0]
0x09	Gyro Y [15:8]
0x0A	Gyro Y [7:0]
0x0B	Gyro Z [15:8]
0x0C	Gyro Z [7:0]
0x0D	Temperature[7:0]
0x0E	TimeStamp[15:8]
0x0F	TimeStamp[7:0]

6.2 FIFO HEADER

The following table shows the structure of the 1-byte FIFO header.

BIT FIELD	ITEM	DESCRIPTION
7	HEADER_MSG	1: FIFO is empty 0: Packet contains sensor data
6	HEADER_ACCEL	1: Packet is sized so that accel data have location in the packet, FIFO_ACCEL_EN must be 1 0: Packet does not contain accel sample
5	HEADER_GYRO	1: Packet is sized so that gyro data have location in the packet, FIFO_GYRO_EN must be 1 0: Packet does not contain gyro sample
3:2	HEADER_TIMESTAMP_FSYNC	00: Packet does not contain timestamp or FSYNC time data 01: Reserved 10: Packet contains ODR Timestamp 11: Packet contains FSYNC time, and this packet is flagged as first ODR after FSYNC (only if FIFO_TMST_FSYNC_EN is 1)
1	HEADER_ODR_ACCEL	1: The ODR for accel is different for this accel data packet compared to the previous accel packet

		0: The ODR for accel is the same as the previous packet with accel
0	HEADER_ODR_GYRO	1: The ODR for gyro is different for this gyro data packet compared to the previous gyro packet 0: The ODR for gyro is the same as the previous packet with gyro

Note at least HEADER_ACCEL or HEADER_GYRO must be set for a sensor data packet to be set.

6.3 MAXIMUM FIFO STORAGE

The maximum number of packets that can be stored in FIFO is a variable quantity depending on the use case. As shown in the figure below, the physical FIFO size is 2048 bytes. A number of bytes equal to the packet size selected (see section 6.1) is reserved to prevent reading a packet during write operation. Additionally, a read cache 2 packets wide is available.

When there is no serial interface operation, the read cache is not available for storing packets, being fed by the serial interface clock.

When serial interface operation happens, depending on the operation length and the packet size chosen, either 1 or 2 of the packet entries in read cache may become available for storing packets. In that case the total storage available is up to the maximum number of packets that can be accommodated in 2048 bytes + 1 packet size, depending on the packet size used.

Due to the non-deterministic nature of system operation, driver memory allocation should always be the largest size of 2080 bytes.

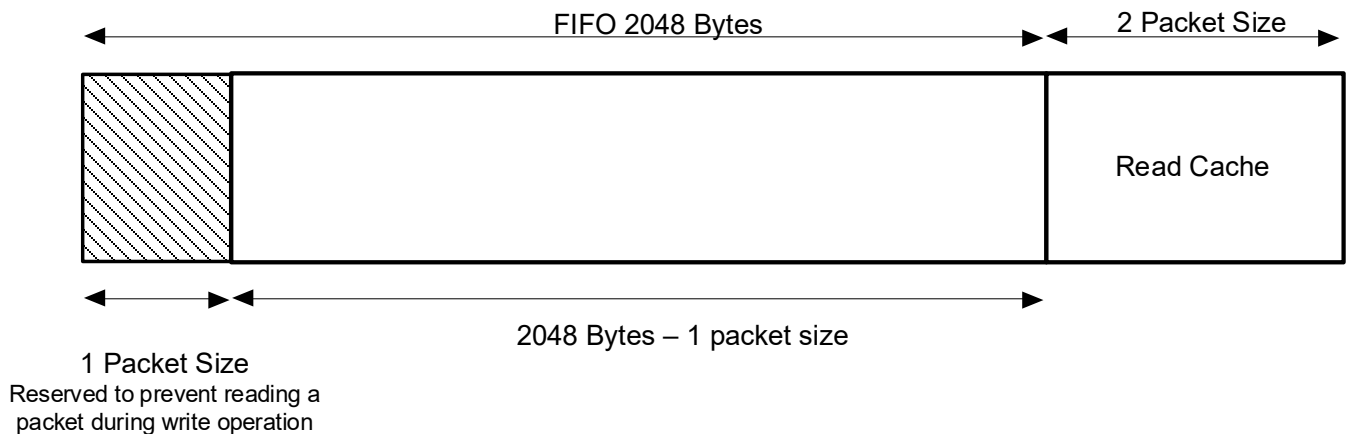


Figure 12. Maximum FIFO Storage

6.4 FIFO CONFIGURATION REGISTERS

The following control bits in bank 0, register 0x5Fh determine what data is placed into the FIFO. The values of these bits may change while the FIFO is being filled without corruption of the FIFO.

BIT	NAME	FUNCTION
3	FIFO_TMST_FSYNC_EN	0: FIFO will only contain ODR timestamp information 1: FIFO can also contain FSYNC time and FSYNC tag for one ODR after an FSYNC event
1	FIFO_GYRO_EN	0: Default setting; Gyroscope data not placed into FIFO 1: Enables gyroscope data packets of 6-bytes to be placed in FIFO
0	FIFO_ACCEL_EN	0: Default setting; Accelerometer data not placed into FIFO 1: Enables accelerometer data packets of 6-bytes to be placed in FIFO

Configuration register settings above impact FIFO header and FIFO packet size as follows:

FIFO_ACCEL_EN	FIFO_GYRO_EN	FIFO_TMST_FSYNC_EN	Header	Packet size
1	1	0	8'b_0110_10xx	16 Bytes
1	1	1	8'b_0110_1xxx	16 Bytes
1	0	X	8'b_0100_00xx	8 Bytes
0	1	X	8'b_0010_00xx	8 Bytes
0	0	X	No FIFO writes	No FIFO writes

7 PROGRAMMABLE INTERRUPTS

The ICM-40609-D has a programmable interrupt system that can generate an interrupt signal on the INT pins. Status flags indicate the source of an interrupt. Interrupt sources may be enabled and disabled individually. There are two interrupt outputs. Any interrupt may be mapped to either interrupt pin as explained in the register section. The following configuration options are available for the interrupts

- INT1 and INT2 can be push-pull or open drain
- Level or pulse mode
- Active high or active low

7.1 WAKE-ON MOTION INTERRUPT

The ICM-40609-D provides motion detection capability. A qualifying motion sample is one where the high passed sample from any axis has an absolute value exceeding a user-programmable threshold. The following steps explain how to configure the Wake-on-Motion Interrupt.

- Wake on Motion configuration parameters
 1. WOM_X_TH (Register 0x4Ah in Bank 4)
 2. WOM_Y_TH (Register 0x4Bh in Bank 4)
 3. WOM_Z_TH (Register 0x4Ch in Bank 4)
 4. WOM_INT_MODE (Register 0x57h in Bank 0)
 5. WOM_MODE (Register 0x57h in Bank 0)
 6. WOM_MODE1 (Register 0x57h in Bank 0)
- Initialize Sensor in a typical configuration
 1. Set accelerometer ODR (Register 0x50h in Bank 0)
ACCEL_ODR = 9 for 50 Hz
 2. Set Accel to Low Power mode (Register 0x4Eh in Bank 0)
ACCEL_MODE = 2 and (Register 0x4Dh in Bank 0), ACCEL_LP_CLK_SEL = 0, for low power mode
 3. Wait 1 millisecond
- Initialize Wake on Motion hardware
 1. Set WOM_X_TH to 98 (Register 0x4Ah in Bank 4)
 2. Set WOM_Y_TH to 98 (Register 0x4Bh in Bank 4)
 3. Set WOM_Z_TH to 98 (Register 0x4Ch in Bank 4)
 4. Wait 1 millisecond
 5. Set WOM_INT_MODE to 0 (Register 0x57h in Bank 0)
 6. Set WOM_MODE to 1 (Register 0x57h in Bank 0)
 7. Set WOM_MODE1 to 1 (Register 0x57h in Bank 0)
 8. Wait 1 millisecond
 9. Enable all 3 axes as WOM sources for INT1 by setting bits 2:0 in register INT_SOURCE1 (Register 0x66h in Bank 0) to 1. Or if INT2 is selected for WOM, enable all 3 axes as WOM sources by setting bits 2:0 in register INT_SOURCE4 (Register 0x69h in Bank 0) to 1.
 10. Wait 50 milliseconds
- Output registers
 1. Read interrupt register (Register 0x37h in Bank 0) for WOM_X_INT
 2. Read interrupt register (Register 0x37h in Bank 0) for WOM_Y_INT
 3. Read interrupt register (Register 0x37h in Bank 0) for WOM_Z_INT

8 DIGITAL INTERFACE

8.1 I²C AND SPI SERIAL INTERFACES

The internal registers and memory of the ICM-40609-D can be accessed using I²C at 1 MHz or SPI at 24 MHz. SPI operates in 3-wire or 4-wire mode. Pin assignments for serial interfaces are described in Section 4.1.

8.2 I²C INTERFACE

I²C is a two-wire interface comprised of the signals serial data (SDA) and serial clock (SCL). In general, the lines are open-drain and bi-directional. In a generalized I²C interface implementation, attached devices can be a master or a slave. The master device puts the slave address on the bus, and the slave device with the matching address acknowledges the master.

The ICM-40609-D always operates as a slave device when communicating to the system processor, which thus acts as the master. SDA and SCL lines typically need pull-up resistors to VDDIO. The maximum bus speed is 1 MHz.

The slave address of the ICM-40609-D is b110100X, which is 7 bits long. The LSB bit of the 7-bit address is determined by the logic level on pin AP_AD0. This allows two ICM-40609-Ds to be connected to the same I²C bus. When used in this configuration, the address of one of the devices should be b1101000 (pin AP_AD0 is logic low) and the address of the other should be b1101001 (pin AP_AD0 is logic high).

8.3 I²C COMMUNICATIONS PROTOCOL

START (S) and STOP (P) Conditions

Communication on the I²C bus starts when the master puts the START condition (S) on the bus, which is defined as a HIGH-to-LOW transition of the SDA line while SCL line is HIGH (see figure below). The bus is considered to be busy until the master puts a STOP condition (P) on the bus, which is defined as a LOW to HIGH transition on the SDA line while SCL is HIGH (see figure below).

Additionally, the bus remains busy if a repeated START (Sr) is generated instead of a STOP condition.

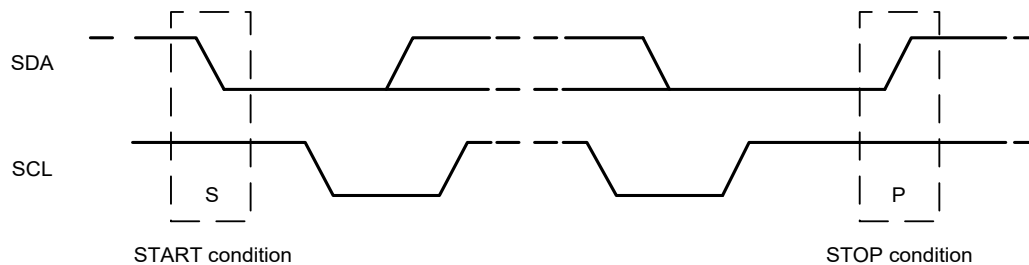


Figure 13. START and STOP Conditions

Data Format / Acknowledge

I²C data bytes are defined to be 8-bits long. There is no restriction to the number of bytes transmitted per data transfer. Each byte transferred must be followed by an acknowledge (ACK) signal. The clock for the acknowledge signal is generated by the master, while the receiver generates the actual acknowledge signal by pulling down SDA and holding it low during the HIGH portion of the acknowledge clock pulse.

If a slave is busy and cannot transmit or receive another byte of data until some other task has been performed, it can hold SCL LOW, thus forcing the master into a wait state. Normal data transfer resumes when the slave is ready, and releases the clock line (refer to the following figure).

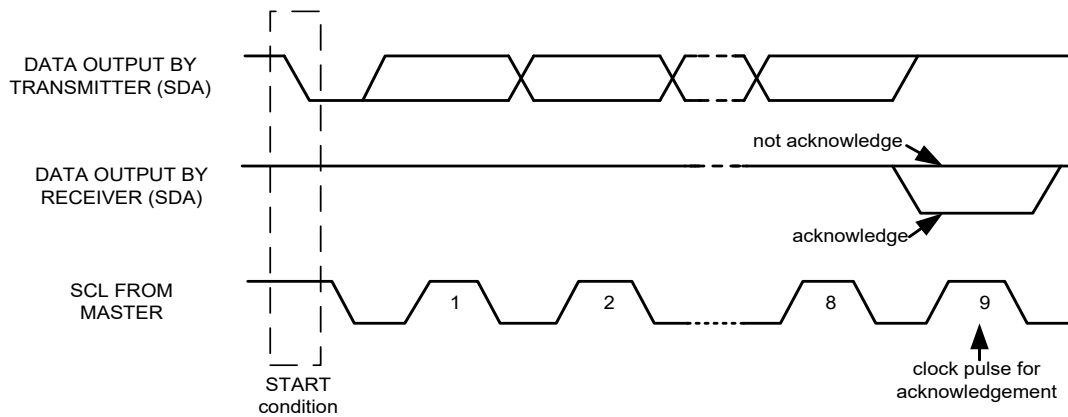


Figure 14. Acknowledge on the I²C Bus

Communications

After beginning communications with the START condition (S), the master sends a 7-bit slave address followed by an 8th bit, the read/write bit. The read/write bit indicates whether the master is receiving data from or is writing to the slave device. Then, the master releases the SDA line and waits for the acknowledge signal (ACK) from the slave device. Each byte transferred must be followed by an acknowledge bit. To acknowledge, the slave device pulls the SDA line LOW and keeps it LOW for the high period of the SCL line. Data transmission is always terminated by the master with a STOP condition (P), thus freeing the communications line. However, the master can generate a repeated START condition (Sr), and address another slave without first generating a STOP condition (P). A LOW to HIGH transition on the SDA line while SCL is HIGH defines the stop condition. All SDA changes should take place when SCL is low, with the exception of start and stop conditions.

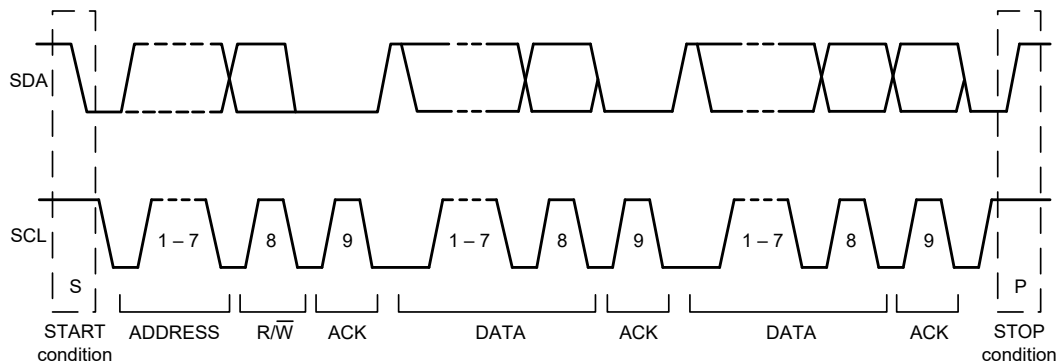


Figure 15. Complete I²C Data Transfer

To write the internal ICM-40609-D registers, the master transmits the start condition (S), followed by the I²C address and the write bit (0). At the 9th clock cycle (when the clock is high), the ICM-40609-D acknowledges the transfer. Then the master puts the register address (RA) on the bus. After the ICM-40609-D acknowledges the reception of the register address, the master puts the register data onto the bus. This is followed by the ACK signal, and data transfer may be concluded by the stop condition (P). To write multiple bytes after the last ACK signal, the master can continue outputting data rather than transmitting a stop signal. In this case, the ICM-40609-D automatically increments the register address and loads the data to the appropriate register. The following figures show single and two-byte write sequences.

Single-Byte Write Sequence

Master	S	AD+W		RA		DATA		P
Slave			ACK		ACK		ACK	

Burst Write Sequence

Master	S	AD+W		RA		DATA		DATA		P
Slave			ACK		ACK		ACK		ACK	

To read the internal ICM-40609-D registers, the master sends a start condition, followed by the I²C address and a write bit, and then the register address that is going to be read. Upon receiving the ACK signal from the ICM-40609-D, the master transmits a start signal followed by the slave address and read bit. As a result, the ICM-40609-D sends an ACK signal and the data. The communication ends with a not acknowledge (NACK) signal and a stop bit from master. The NACK condition is defined such that the SDA line remains high at the 9th clock cycle. The following figures show single- and two-byte read sequences.

Single-Byte Read Sequence

Master	S	AD+W		RA		S	AD+R			NACK	P
Slave			ACK		ACK			ACK	DATA		

Burst Read Sequence

Master	S	AD+W		RA		S	AD+R			ACK		NACK	P
Slave			ACK		ACK			ACK	DATA		DATA		

8.4 I²C TERMS

Signal	Description
S	Start Condition: SDA goes from high to low while SCL is high
AD	Slave I ² C address
W	Write bit (0)
R	Read bit (1)
ACK	Acknowledge: SDA line is low while the SCL line is high at the 9 th clock cycle
NACK	Not-Acknowledge: SDA line stays high at the 9 th clock cycle
RA	ICM-40609-D internal register address
DATA	Transmit or received data
P	Stop condition: SDA going from low to high while SCL is high

Table 12. I²C Terms

8.5 SPI INTERFACE

The ICM-40609-D supports 3-wire or 4-wire SPI for the host interface. The ICM-40609-D always operates as a Slave device during standard Master-Slave SPI operation.

With respect to the Master, the Serial Clock output (SCLK), the Serial Data Output (SDO), the Serial Data Input (SDI), and the Serial Data IO (SDIO) are shared among the Slave devices. Each SPI slave device requires its own Chip Select (CS) line from the master.

CS goes low (active) at the start of transmission and goes back high (inactive) at the end. Only one CS line is active at a time, ensuring that only one slave is selected at any given time. The CS lines of the non-selected slave devices are held high, causing their SDO lines to remain in a high-impedance (high-z) state so that they do not interfere with any active devices.

SPI Operational Features

1. Data is delivered MSB first and LSB last
2. Data is latched on the rising edge of SCLK
3. Data should be transitioned on the falling edge of SCLK
4. The maximum frequency of SCLK is 24 MHz
5. SPI read and write operations are completed in 16 or more clock cycles (two or more bytes). The first byte contains the SPI Address, and the following byte(s) contain(s) the SPI data. The first bit of the first byte contains the Read/Write bit and indicates the Read (1) or Write (0) operation. The following 7 bits contain the Register Address. In cases of multiple-byte Read/Writes, data is two or more bytes:

SPI Address format

MSB							LSB
R/W	A6	A5	A4	A3	A2	A1	A0

SPI Data format

MSB							LSB
D7	D6	D5	D4	D3	D2	D1	D0

6. Supports Single or Burst Read/Writes.

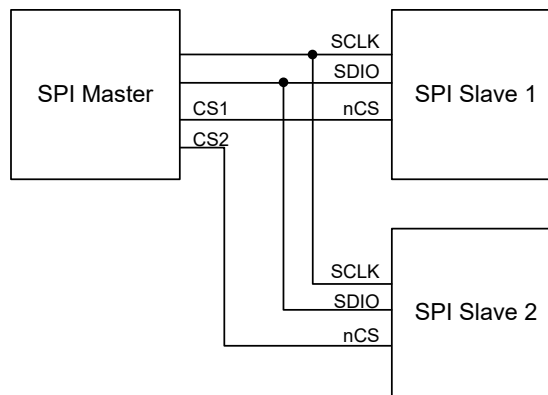


Figure 16. Typical SPI Master/Slave Configuration

9 ASSEMBLY

This section provides general guidelines for assembling InvenSense Micro Electro-Mechanical Systems (MEMS) gyros packaged in LGA package.

9.1 ORIENTATION OF AXES

The diagram below shows the orientation of the axes of sensitivity and the polarity of rotation. Note the pin 1 identifier (•) in the figure.

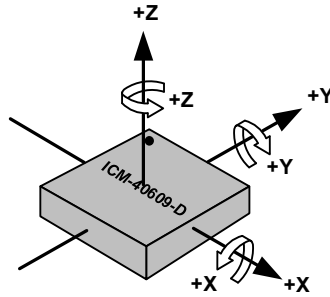
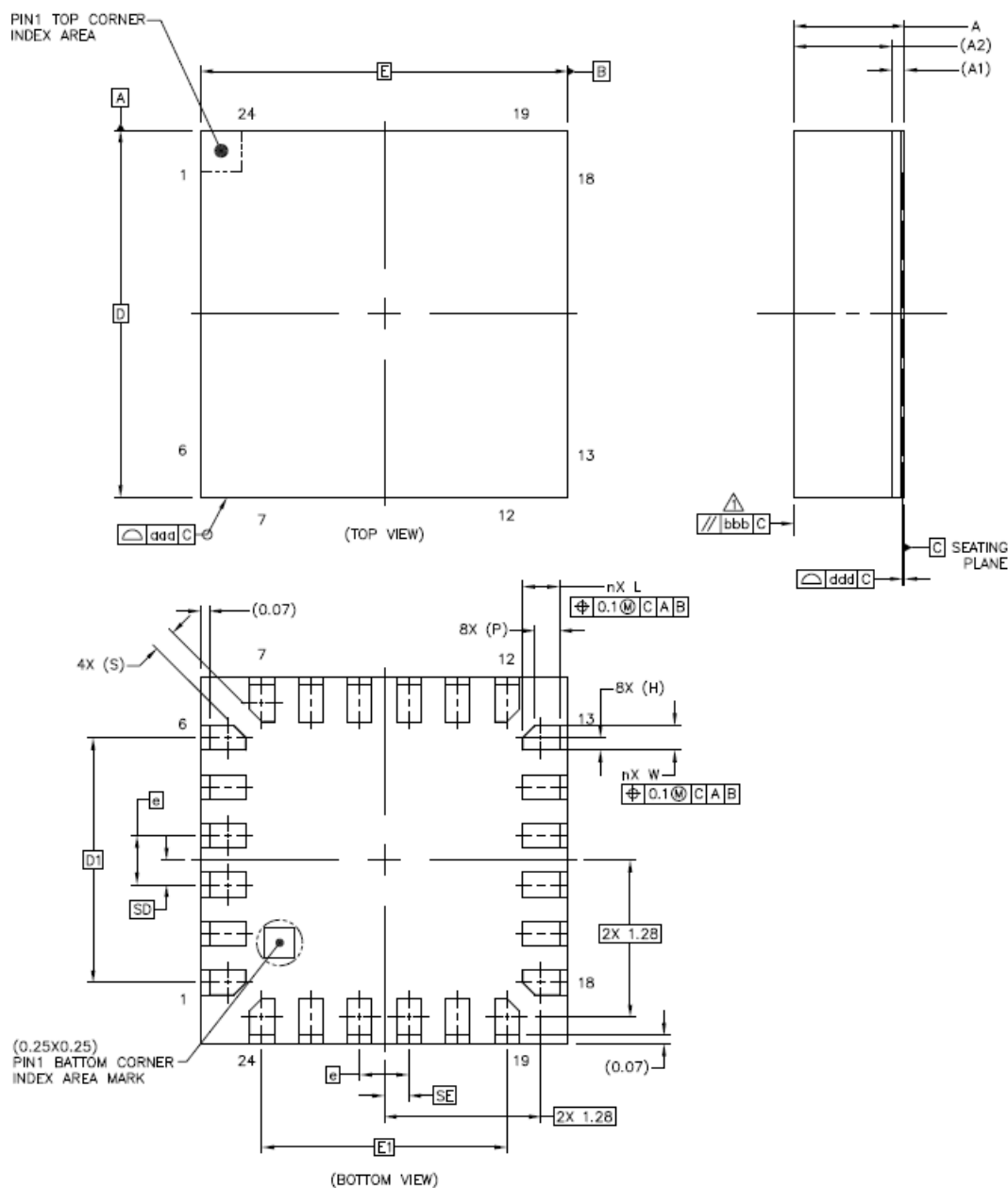


Figure 17. Orientation of Axes of Sensitivity and Polarity of Rotation

9.2 PACKAGE DIMENSIONS

24 Lead LGA (3x3x0.91) mm NiAu pad finish

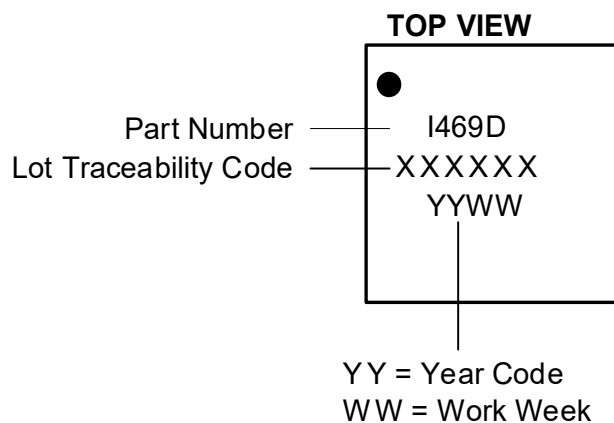


SYMBOLS	DIMENSIONS IN MILLIMETERS		
	MIN	NOM	MAX
A	0.85	0.91	0.97
A1	0.105		REF
A2	0.8		REF
D	2.9	3	3.1
E	2.9	3	3.1
W	0.15	0.2	0.25
L	0.25	0.3	0.35
e	0.4		BSC
n	24		
D1	2		BSC
E1	2		BSC
SD	0.2		BSC
SE	0.2		BSC
b	---	---	---
e1	---		
n1	---		
aaa	0.1		
bbb	0.2		
ddd	0.08		
eee	---		
fff	---		
H		0.1	REF
P		0.2	REF
S		0.184	REF

10 PART NUMBER PACKAGE MARKING

The part number package marking for ICM-40609-D devices is summarized below:

Part Number	Part Number Package Marking
ICM-40609-D	I469D



11 USE NOTES

11.1 ACCELEROMETER MODE TRANSITIONS

When transitioning from accelerometer Low Power (LP) mode to accelerometer Low Noise (LN) mode, if ODR is 6.25 Hz or lower, software should change ODR to a value of 12.5 Hz or higher, because accelerometer LN mode does not support ODR values below 12.5 Hz.

When transitioning from accelerometer LN mode to accelerometer LP mode, if ODR is greater than 500 Hz, software should change ODR to a value of 500Hz or lower, because accelerometer LP mode does not support ODR values above 500 Hz.

11.2 ACCELEROMETER LOW POWER (LP) MODE AVERAGING FILTER SETTING

Software drivers provided with the device use Averaging Filter setting of 16x. This setting is recommended for meeting Android noise requirements in LP mode, and to minimize accelerometer offset variation when transitioning from LP to Low Noise (LN) mode. 1x averaging filter can be used by following the setting configuration shown in section 14.38.

11.3 SETTINGS FOR I²C, AND SPI OPERATION

Upon bootup the device comes up in SPI mode. The following settings should be used for I²C, and SPI operation.

Register Field	I ² C Driver Setting	SPI Driver Setting
I2C_SLEW_RATE (bits 5:3, register DRIVE_CONFIG, address 0x13, bank 0)	1	0
SPI_SLEW_RATE (bits 2:0, register DRIVE_CONFIG, address 0x13, bank 0)	1	5

11.4 NOTCH FILTER AND ANTI-ALIAS FILTER OPERATION

Use of Notch Filter and Anti-Alias Filter is supported only for Low Noise (LN) mode operation. The host is responsible for keeping the UI path in LN mode while Notch Filter and Anti-Alias Filter are turned on.

11.5 REGISTER VALUES MODIFICATION

The only register settings that user can modify during sensor operation are for ODR selection, FSR selection, and sensor mode changes (register parameters GYRO_ODR, ACCEL_ODR, GYRO_FS_SEL, ACCEL_FS_SEL, GYRO_MODE, ACCEL_MODE). User must not modify any other register values during sensor operation. The following procedure must be used for other register values modification.

- Turn Accel and Gyro Off
- Modify register values
- Turn Accel and/or Gyro On

12 REGISTER MAP

This section lists the register map for the ICM-40609-D, for user banks 0, 1, 2, 4.

12.1 USER BANK 0 REGISTER MAP

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
11	17	DEVICE_CONFIG	R/W	-			SPI_MODE	-			SOFT_RESET_CONFIG
13	19	DRIVE_CONFIG	R/W	-		I2C_SLEW_RATE			SPI_SLEW_RATE		
14	20	INT_CONFIG	R/W	-		INT2_MODE	INT2_DRIVE_CIRCUIT	INT2_POLARITY	INT1_MODE	INT1_DRIVE_CIRCUIT	INT1_POLARITY
16	22	FIFO_CONFIG	R/W	FIFO_MODE		-					
1D	29	TEMP_DATA1	SYNCR	TEMP_DATA[15:8]							
1E	30	TEMP_DATA0	SYNCR	TEMP_DATA[7:0]							
1F	31	ACCEL_DATA_X1	SYNCR	ACCEL_DATA_X[15:8]							
20	32	ACCEL_DATA_X0	SYNCR	ACCEL_DATA_X[7:0]							
21	33	ACCEL_DATA_Y1	SYNCR	ACCEL_DATA_Y[15:8]							
22	34	ACCEL_DATA_Y0	SYNCR	ACCEL_DATA_Y[7:0]							
23	35	ACCEL_DATA_Z1	SYNCR	ACCEL_DATA_Z[15:8]							
24	36	ACCEL_DATA_Z0	SYNCR	ACCEL_DATA_Z[7:0]							
25	37	GYRO_DATA_X1	SYNCR	GYRO_DATA_X[15:8]							
26	38	GYRO_DATA_X0	SYNCR	GYRO_DATA_X[7:0]							
27	39	GYRO_DATA_Y1	SYNCR	GYRO_DATA_Y[15:8]							
28	40	GYRO_DATA_Y0	SYNCR	GYRO_DATA_Y[7:0]							
29	41	GYRO_DATA_Z1	SYNCR	GYRO_DATA_Z[15:8]							
2A	42	GYRO_DATA_Z0	SYNCR	GYRO_DATA_Z[7:0]							
2B	43	TMST_FSYNCH	SYNCR	TMST_FSYNC_DATA[15:8]							
2C	44	TMST_FSYNCL	SYNCR	TMST_FSYNC_DATA[7:0]							
2D	45	INT_STATUS	R/C		UI_FSYNC_INT	PLL_RDY_INT	RESET_DONE_INT	DATA_RDY_INT	FIFO_THS_INT	FIFO_FULL_INT	AGC_RDY_INT
2E	46	FIFO_COUNTH	R	FIFO_COUNT[15:8]							
2F	47	FIFO_COUNTL	R	FIFO_COUNT[7:0]							
30	48	FIFO_DATA	R	FIFO_DATA							
37	55	INT_STATUS2	R/C	-					WOM_Z_INT	WOM_Y_INT	WOM_X_INT
4B	75	SIGNAL_PATH_RESET	W/C	-				ABORT_AND_RESET	TMST_STROBE	FIFO_FLUSH	-
4C	76	INTF_CONFIG0	R/W	FIFO_HOLD_LAST_DATA_EN	FIFO_COUNT_REC	FIFO_COUNT_ENDIAN	SENSOR_DATA_ENDIAN	-		UI_SIFS_CFG	
4D	77	INTF_CONFIG1	R/W	EN_TEST_MODE		-		ACCEL_LP_CLK_SEL	-	CLKSEL	
4E	78	PWR_MGMT0	R/W	-		TEMP_DIS	IDLE	GYRO_MODE		ACCEL_MODE	
4F	79	GYRO_CONFIG0	R/W	GYRO_FS_SEL			-	GYRO_ODR			
50	80	ACCEL_CONFIG0	R/W	ACCEL_FS_SEL			-	ACCEL_ODR			
51	81	GYRO_CONFIG1	R/W	TEMP_FILT_BW			-	GYRO_UI_FILT_ORD		GYRO_DEC2_M2_ORD	
52	82	GYRO_ACCEL_CONFIG0	R/W	ACCEL_UI_FILT_BW				GYRO_UI_FILT_BW			
53	83	ACCEL_CONFIG1	R/W	-			ACCEL_UI_FILT_ORD		ACCEL_DEC2_M2_ORD		-
54	84	TMST_CONFIG	R/W	-			TMST_TO_REGS_EN	TMST_RES	TMST_DELTA_EN	TMST_FSYNC_EN	TMST_EN
57	87	WOM_CONFIG	R/W	-				WOM_INT_MODE	WOM_MODE	WOM_MODE1	
5F	95	FIFO_CONFIG1	R/W	-	FIFO_RESUME_PARTIAL_READ	FIFO_WM_GTT_TH	-	FIFO_TMST_FSYNC_EN	FIFO_TEMP_EN	FIFO_GYRO_EN	FIFO_ACCEL_EN
60	96	FIFO_CONFIG2	R/W	FIFO_WM[7:0]							
61	97	FIFO_CONFIG3	R/W	-				FIFO_WM[11:8]			

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
62	98	FSYNC_CONFIG	R/W	-	FSYNC_UI_SEL			-		FSYNC_UI_FL AG_CLEAR_S EL	FSYNC_POLA RITY
63	99	INT_CONFIG0	R/W	-		UI_DRDY_INT_CLEAR		FIFO_THS_INT_CLEAR		FIFO_FULL_INT_CLEAR	
64	100	INT_CONFIG1	R/W	-	INT_TPULSE_ DURATION	INT_TDEASSE RT_DISABLE	INT_ASYNC_ RESET	-			
65	101	INT_SOURCE0	R/W	-	UI_FSYNC_IN T1_EN	PLL_RDY_INT 1_EN	RESET_DONE _INT1_EN	UI_DRDY_INT 1_EN	FIFO_THS_IN T1_EN	FIFO_FULL_I NT1_EN	UI_AGC_RDY _INT1_EN
66	102	INT_SOURCE1	R/W	-					WOM_Z_INT 1_EN	WOM_Y_INT 1_EN	WOM_X_INT 1_EN
68	104	INT_SOURCE3	R/W	-	UI_FSYNC_IN T2_EN	PLL_RDY_INT 2_EN	RESET_DONE _INT2_EN	UI_DRDY_INT 2_EN	FIFO_THS_IN T2_EN	FIFO_FULL_I NT2_EN	UI_AGC_RDY _INT2_EN
69	105	INT_SOURCE4	R/W	-					WOM_Z_INT 2_EN	WOM_Y_INT 2_EN	WOM_X_INT 2_EN
6C	108	FIFO_LOST_PKT0	R	FIFO_LOST_PKT_CNT[15:8]							
6D	109	FIFO_LOST_PKT1	R	FIFO_LOST_PKT_CNT[7:0]							
70	112	SELF_TEST_CONFIG	R/W		ACCEL_ST_P OWER	EN_AZ_ST	EN_AY_ST	EN_AX_ST	EN_GZ_ST	EN_GY_ST	EN_GX_ST
75	117	WHO_AM_I	R	WHOAMI							
76	118	REG_BANK_SEL	R/W	-					BANK_SEL		

12.2 USER BANK 1 REGISTER MAP

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	
03	03	SENSOR_CONFIG0	R/W	-		ZG_DISABLE	YG_DISABLE	XG_DISABLE	ZA_DISABLE	YA_DISABLE	XA_DISABLE	
0B	11	GYRO_CONFIG_STATIC2	R/W	-							GYRO_AAF_D IS	GYRO_NF_DI S
0C	12	GYRO_CONFIG_STATIC3	R/W	-		GYRO_AAF_DELT						
0D	13	GYRO_CONFIG_STATIC4	R/W	GYRO_AAF_DELTSQR[7:0]								
0E	14	GYRO_CONFIG_STATIC5	R/W	GYRO_AAF_BITSHIFT					GYRO_AAF_DELTSQR[11:8]			
0F	15	GYRO_CONFIG_STATIC6	R/W	GYRO_X_NF_COSWZ[7:0]								
10	16	GYRO_CONFIG_STATIC7	R/W	GYRO_Y_NF_COSWZ[7:0]								
11	17	GYRO_CONFIG_STATIC8	R/W	GYRO_Z_NF_COSWZ[7:0]								
12	18	GYRO_CONFIG_STATIC9	R/W	-		GYRO_Z_NF_COSWZ_SEL[0]	GYRO_Y_NF_COSWZ_SEL[0]	GYRO_X_NF_COSWZ_SEL[0]	GYRO_Z_NF_COSWZ[8]	GYRO_Y_NF_COSWZ[8]	GYRO_X_NF_COSWZ[8]	
13	19	GYRO_CONFIG_STATIC10	R/W	-	GYRO_NF_BW_SEL				GYRO_HPF_BW_IND			GYRO_HPF_O RD_IND
5F	95	XG_ST_DATA	R/W	XG_ST_DATA								
60	96	YG_ST_DATA	R/W	YG_ST_DATA								
61	97	ZG_ST_DATA	R/W	ZG_ST_DATA								
62	98	TMSTVAL0	R	TMST_VALUE[7:0]								
63	99	TMSTVAL1	R	TMST_VALUE[15:8]								
64	100	TMSTVAL2	R	-					TMST_VALUE[19:16]			
7A	122	INTF_CONFIG4	R/W	-							SPI_AP_4WIR E	-
7B	123	INTF_CONFIG5	R/W	-					PIN19_FUNCTION			-

12.3 USER BANK 2 REGISTER MAP

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
03	03	ACCEL_CONFIG_STATIC2	R/W	-	ACCEL_AAF_DELT						ACCEL_AAF_DIS
04	04	ACCEL_CONFIG_STATIC3	R/W	ACCEL_AAF_DELTSQR[7:0]							
05	05	ACCEL_CONFIG_STATIC4	R/W	ACCEL_AAF_BITSHIFT				ACCEL_AAF_DELTSQR[11:8]			
3B	59	XA_ST_DATA	R/W	XA_ST_DATA							
3C	60	YA_ST_DATA	R/W	YA_ST_DATA							
3D	61	ZA_ST_DATA	R/W	ZA_ST_DATA							

12.4 USER BANK 4 REGISTER MAP

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
4A	74	ACCEL_WOM_X_THR	R/W	WOM_X_TH							
4B	75	ACCEL_WOM_Y_THR	R/W	WOM_Y_TH							
4C	76	ACCEL_WOM_Z_THR	R/W	WOM_Z_TH							
77	119	OFFSET_USER0	R/W	GYRO_X_OFFUSER[7:0]							
78	120	OFFSET_USER1	R/W	GYRO_Y_OFFUSER[11:8]				GYRO_X_OFFUSER[11:8]			
79	121	OFFSET_USER2	R/W	GYRO_Y_OFFUSER[7:0]							
7A	122	OFFSET_USER3	R/W	GYRO_Z_OFFUSER[7:0]							
7B	123	OFFSET_USER4	R/W	ACCEL_X_OFFUSER[11:8]				GYRO_Z_OFFUSER[11:8]			
7C	124	OFFSET_USER5	R/W	ACCEL_X_OFFUSER[7:0]							
7D	125	OFFSET_USER6	R/W	ACCEL_Y_OFFUSER[7:0]							
7E	126	OFFSET_USER7	R/W	ACCEL_Z_OFFUSER[11:8]				ACCEL_Y_OFFUSER[11:8]			
7F	127	OFFSET_USER8	R/W	ACCEL_Z_OFFUSER[7:0]							

Detailed register descriptions are provided in the sections that follow. Please note the following regarding Clock Domain for each register:

1. Clock Domain: SCLK_UI means that the register is controlled from the UI interface

Register fields marked as Reserved must not be modified by the user. The Reset Value of the register can be used to determine the default value of reserved register fields, and unless otherwise noted this default value must be maintained even if the values of other register fields are modified by the user.

13 USER BANK 0 REGISTER MAP – DESCRIPTIONS

This section describes the function and contents of each register within USR Bank 0.

Note: The device powers up in sleep mode.

13.1 DEVICE_CONFIG

Name: DEVICE_CONFIG Address: 17 (11h) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:5	-	Reserved
4	SPI_MODE	SPI mode selection 0: Mode 0 and Mode 3 (default) 1: Mode 1 and Mode 2
3:1	-	Reserved
0	SOFT_RESET_CONFIG	Software reset configuration 0: Normal (default) 1: Enable reset After writing 1 to this bitfield, wait 1ms for soft reset to be effective, before attempting any other register access

13.2 DRIVE_CONFIG

Name: DRIVE_CONFIG Address: 19 (13h) Serial IF: R/W Reset value: 0x05 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:6	-	Reserved
5:3	I2C_SLEW_RATE	Controls slew rate for output pin 14 in I ² C mode only 000: 20ns-60ns 001: 12ns-36ns 010: 6ns-18ns 011: 4ns-12ns 100: 2ns-6ns 101: < 2ns 110: Reserved 111: Reserved
2:0	SPI_SLEW_RATE	Controls slew rate for output pin 14 in SPI mode, and for all other output pins 000: 20ns-60ns 001: 12ns-36ns 010: 6ns-18ns 011: 4ns-12ns 100: 2ns-6ns 101: < 2ns 110: Reserved 111: Reserved

13.3 INT_CONFIG

Name: INT_CONFIG Address: 20 (14h) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_OIS1		
BIT	NAME	FUNCTION
7:6	-	Reserved
5	INT2_MODE	INT2 interrupt mode 0: Pulsed mode 1: Latched mode
4	INT2_DRIVE_CIRCUIT	INT2 drive circuit 0: Open drain 1: Push pull
3	INT2_POLARITY	INT2 interrupt polarity 0: Active low (default) 1: Active high
2	INT1_MODE	INT1 interrupt mode 0: Pulsed mode 1: Latched mode
1	INT1_DRIVE_CIRCUIT	INT1 drive circuit 0: Open drain 1: Push pull
0	INT1_POLARITY	INT1 interrupt polarity 0: Active low (default) 1: Active high

13.4 FIFO_CONFIG

Name: FIFO_CONFIG Address: 22 (16h) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:6	FIFO_MODE	00: Bypass Mode (default) 01: Stream-to-FIFO Mode 10: STOP-on-FULL Mode 11: STOP-on-FULL Mode
5:0	-	Reserved

13.5 TEMP_DATA1

Name: TEMP_DATA1 Address: 29 (1Dh) Serial IF: SYNCR Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	TEMP_DATA[15:8]	Upper byte of temperature data

13.6 TEMP_DATA0

Name: TEMP_DATA0
Address: 30 (1Eh)
Serial IF: SYNCR
Reset value: 0x00
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	TEMP_DATA[7:0]	Lower byte of temperature data

Temperature data value from the sensor data registers can be converted to degrees centigrade by using the following formula:

$$\text{Temperature in Degrees Centigrade} = (\text{TEMP_DATA} / 132.48) + 25$$

Temperature data stored in FIFO is an 8-bit quantity, FIFO_TEMP_DATA. It can be converted to degrees centigrade by using the following formula:

$$\text{Temperature in Degrees Centigrade} = (\text{FIFO_TEMP_DATA} / 2.07) + 25$$

13.7 ACCEL_DATA_X1

Name: ACCEL_DATA_X1
Address: 31 (1Fh)
Serial IF: SYNCR
Reset value: 0x00
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	ACCEL_DATA_X[15:8]	Upper byte of Accel X-axis data

13.8 ACCEL_DATA_X0

Name: ACCEL_DATA_X0
Address: 32 (20h)
Serial IF: SYNCR
Reset value: 0x00
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	ACCEL_DATA_X[7:0]	Lower byte of Accel X-axis data

13.9 ACCEL_DATA_Y1

Name: ACCEL_DATA_Y1
Address: 33 (21h)
Serial IF: SYNCR
Reset value: 0x00
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	ACCEL_DATA_Y[15:8]	Upper byte of Accel Y-axis data

13.10 ACCEL_DATA_Y0

Name: ACCEL_DATA_Y0
Address: 34 (22h)
Serial IF: SYNCR
Reset value: 0x00
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	ACCEL_DATA_Y[7:0]	Lower byte of Accel Y-axis data

13.11 ACCEL_DATA_Z1

Name: ACCEL_DATA_Z1
 Address: 35 (23h)
 Serial IF: SYNCR
 Reset value: 0x00
 Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	ACCEL_DATA_Z[15:8]	Upper byte of Accel Z-axis data

13.12 ACCEL_DATA_Z0

Name: ACCEL_DATA_Z0
 Address: 36 (24h)
 Serial IF: SYNCR
 Reset value: 0x00
 Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	ACCEL_DATA_Z[7:0]	Lower byte of Accel Z-axis data

13.13 GYRO_DATA_X1

Name: GYRO_DATA_X1
 Address: 37 (25h)
 Serial IF: SYNCR
 Reset value: 0x00
 Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	GYRO_DATA_X[15:8]	Upper byte of Gyro X-axis data

13.14 GYRO_DATA_X0

Name: GYRO_DATA_X0
 Address: 38 (26h)
 Serial IF: SYNCR
 Reset value: 0x00
 Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	GYRO_DATA_X[7:0]	Lower byte of Gyro X-axis data

13.15 GYRO_DATA_Y1

Name: GYRO_DATA_Y1
 Address: 39 (27h)
 Serial IF: SYNCR
 Reset value: 0x00
 Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	GYRO_DATA_Y[15:8]	Upper byte of Gyro Y-axis data

13.16 GYRO_DATA_Y0

Name: GYRO_DATA_Y0
Address: 40 (28h)
Serial IF: SYNCR
Reset value: 0x00
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	GYRO_DATA_Y[7:0]	Lower byte of Gyro Y-axis data

13.17 GYRO_DATA_Z1

Name: GYRO_DATA_Z1
Address: 41 (29h)
Serial IF: SYNCR
Reset value: 0x00
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	GYRO_DATA_Z[15:8]	Upper byte of Gyro Z-axis data

13.18 GYRO_DATA_Z0

Name: GYRO_DATA_Z0
Address: 42 (2Ah)
Serial IF: SYNCR
Reset value: 0x00
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	GYRO_DATA_Z[7:0]	Lower byte of Gyro Z-axis data

13.19 TMST_FSYNCH

Name: TMST_FSYNCH
Address: 43 (2Bh)
Serial IF: SYNCR
Reset value: 0x00
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	TMST_FSYNC_DATA[15:8]	Stores the upper byte of the time delta from the rising edge of FSYNC to the latest ODR until the UI Interface reads the FSYNC tag in the status register

13.20 TMST_FSYNCL

Name: TMST_FSYNCL
Address: 44 (2Ch)
Serial IF: SYNCR
Reset value: 0x00
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	TMST_FSYNC_DATA[7:0]	Stores the lower byte of the time delta from the rising edge of FSYNC to the latest ODR until the UI Interface reads the FSYNC tag in the status register

13.21 INT_STATUS

Name: INT_STATUS Address: 45 (2Dh) Serial IF: R/C Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7	-	Reserved
6	UI_FSYNC_INT	This bit automatically sets to 1 when a UI FSYNC interrupt is generated. The bit clears to 0 after the register has been read.
5	PLL_RDY_INT	This bit automatically sets to 1 when a PLL Ready interrupt is generated. The bit clears to 0 after the register has been read.
4	RESET_DONE_INT	This bit automatically sets to 1 when software reset is complete. The bit clears to 0 after the register has been read.
3	DATA_RDY_INT	This bit automatically sets to 1 when a Data Ready interrupt is generated. The bit clears to 0 after the register has been read.
2	FIFO_THS_INT	This bit automatically sets to 1 when the FIFO buffer reaches the threshold value. The bit clears to 0 after the register has been read.
1	FIFO_FULL_INT	This bit automatically sets to 1 when the FIFO buffer is full. The bit clears to 0 after the register has been read.
0	AGC_RDY_INT	This bit automatically sets to 1 when an AGC Ready interrupt is generated. The bit clears to 0 after the register has been read.

13.22 FIFO_COUNTH

Name: FIFO_COUNTH Address: 46 (2Eh) Serial IF: R Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	FIFO_COUNT[15:8]	High Bits, count indicates the number of records or bytes available in FIFO according to FIFO_COUNT_REC setting. Note: Must read FIFO_COUNTL to latch new data for both FIFO_COUNTH and FIFO_COUNTL.

13.23 FIFO_COUNTL

Name: FIFO_COUNTL Address: 47 (2Fh) Serial IF: R Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	FIFO_COUNT[7:0]	Low Bits, count indicates the number of records or bytes available in FIFO according to FIFO_COUNT_REC setting. Reading this byte latches the data for both FIFO_COUNTH, and FIFO_COUNTL.

13.24 FIFO_DATA

Name: FIFO_DATA Address: 48 (30h) Serial IF: R Reset value: 0xFF Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	FIFO_DATA	FIFO data port

13.25 INT_STATUS2

Name: INT_STATUS2 Address: 55 (37h) Serial IF: R/C Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:3	-	Reserved
2	WOM_Z_INT	Wake on Motion Interrupt on Z-axis, clears on read
1	WOM_Y_INT	Wake on Motion Interrupt on Y-axis, clears on read
0	WOM_X_INT	Wake on Motion Interrupt on X-axis, clears on read

13.26 SIGNAL_PATH_RESET

Name: SIGNAL_PATH_RESET Address: 75 (4Bh) Serial IF: W/C Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:4	-	Reserved
3	ABORT_AND_RESET	When this bit is set to 1, the signal path is reset by restarting the ODR counter and signal path controls
2	TMST_STROBE	When this bit is set to 1, the time stamp counter is latched into the time stamp register. This is a write on clear bit.
1	FIFO_FLUSH	When set to 1, FIFO will get flushed.
0	-	Reserved

13.27 INTF_CONFIG0

Name: INTF_CONFIG0 Address: 76 (4Ch) Serial IF: R/W Reset value: 0x30 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7	FIFO_HOLD_LAST_DATA_EN	Setting 0 corresponds to the following: Sense Registers from Power on Reset till first sample: Invalid Samples Value: -32768 Sense Registers after first sample received:

		• Sense Registers Valid Sample Values: ○ Range limited from -32766 to +32767 when FSYNC tag is disabled, or for sensor not selected for FSYNC tag ○ Range limited from -32765 to +32767 (odd values) for sensor selected for FSYNC tag, and FSYNC is tagged ○ Range limited from -32766 to +32766 (even values) for sensor selected for FSYNC tag, but FSYNC is not tagged • Sense Registers Invalid Sample Values: ○ -32768 when FSYNC tag is disabled, or for sensor not selected for FSYNC tag, or for sensor selected for FSYNC tag but FSYNC is not tagged ○ -32767 for sensor selected for FSYNC tag, and FSYNC is tagged FIFO: • Invalid Sample Value: -32768 • Valid Sample Values: -32766 to +32767 Setting 1 corresponds to the following: Sense Registers from Power on Reset till first sample: • Invalid Samples Value: 0 Sense Registers after first sample received: • Sense Registers Valid Sample Values: ○ Range limited from -32768 to +32767 when FSYNC tag is disabled, or for sensor not selected for FSYNC tag ○ Range limited from -32767 to +32767 (odd values) for sensor selected for FSYNC tag, and FSYNC is tagged ○ Range limited from -32768 to +32766 (even values) for sensor selected for FSYNC tag, but FSYNC is not tagged • Sense Registers Invalid Sample Values: ○ Registers hold last valid sample until new one arrives FIFO: • Invalid Sample Value: Copy last valid sample • Valid Sample Values: -32768 to +32767
6	FIFO_COUNT_REC	0: FIFO count is reported in bytes 1: FIFO count is reported in records (1 record = 16 bytes for header + gyro + accel + temp sensor data + time stamp, or 8 bytes for header + gyro/accel + temp sensor data)
5	FIFO_COUNT_ENDIAN	0: FIFO count is reported in Little Endian format 1: FIFO count is reported in Big Endian format (default)
4	SENSOR_DATA_ENDIAN	0: Sensor data is reported in Little Endian format 1: Sensor data is reported in Big Endian format (default)
3:2	-	Reserved
1:0	UI_SIFS_CFG	0x: Reserved

		10: Disable SPI 11: Disable I2C
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Invalid Data Generation: FIFO/Sense Registers may contain invalid data under the following conditions:

- From power on reset to first ODR sample of any sensor (accel, gyro, temp sensor)
- When any sensor is disabled (accel, gyro, temp sensor)
- When accel and gyro are enabled with different ODRs. In this case, the sensor with lower ODR will generate invalid samples when it has no new data.

Invalid data can take special values or can hold last valid sample received. For -32768 to be used as a flag for invalid accel/gyro samples, the valid accel/gyro sample range is limited in such case as well. Bit 7 of INTF_CONFIG0 controls what values invalid (and valid) samples can take as shown above.

13.28 INTF_CONFIG1

Name: INTF_CONFIG1 Address: 77 (4Dh) Serial IF: R/W Reset value: 0x91 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:6	EN_TEST_MODE	Enables test mode. Must be set to 01 for normal device operation.
5:4	-	Reserved
3	ACCEL_LP_CLK_SEL	0: Accelerometer LP mode uses Wake Up oscillator clock 1: Accelerometer LP mode uses RC oscillator clock
2	-	Reserved
1:0	CLKSEL	00: Always select internal RC oscillator 01: Select PLL when available, else select RC oscillator (default) 10: Reserved 11: Disable all clocks

13.29 PWR_MGMT0

Name: PWR_MGMT0 Address: 78 (4Eh) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:6	-	Reserved
5	TEMP_DIS	0: Temperature sensor is enabled (default) 1: Temperature sensor is disabled
4	IDLE	If this bit is set to 1, the RC oscillator is powered on even if Accel and Gyro are powered off. Nominally this bit is set to 0, so when Accel and Gyro are powered off, the chip will go to OFF state, since the RC oscillator will also be powered off
3:2	GYRO_MODE	00: Turns gyroscope off (default) 01: Places gyroscope in Standby Mode 10: Reserved 11: Places gyroscope in Low Noise (LN) Mode Gyroscope needs to be kept ON for a minimum of 45ms. When transitioning from OFF to any of the other modes, do not issue any register writes for 200µs.

1:0	ACCEL_MODE	00: Turns accelerometer off (default) 01: Turns accelerometer off 10: Places accelerometer in Low Power (LP) Mode 11: Places accelerometer in Low Noise (LN) Mode When transitioning from OFF to any of the other modes, do not issue any register writes for 200μs.
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13.30 GYRO_CONFIG0

Name: GYRO_CONFIG0 Address: 79 (4Fh) Serial IF: R/W Reset value: 0x07 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:5	GYRO_FS_SEL	Full scale select for gyroscope UI interface output 000: ±2000dps (default) 001: ±1000dps 010: ±500dps 011: ±250dps 100: ±125dps 101: ±62.5dps 110: ±31.25dps 111: ±15.625dps
4	-	Reserved
3:0	GYRO_ODR	Gyroscope ODR selection for UI interface output 0000: Reserved 0001: 32kHz 0010: 16kHz 0011: 8kHz 0100: 4kHz 0101: 2kHz 0110: 1kHz (default) 0111: 200Hz 1000: 100Hz 1001: 50Hz 1010: 25Hz 1011: 12.5Hz 1100: Reserved 1101: Reserved 1110: Reserved 1111: 500Hz

13.31 ACCEL_CONFIG0

Name: ACCEL_CONFIG0
 Address: 80 (50h)
 Serial IF: R/W
 Reset value: 0x07
 Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:5	ACCEL_FS_SEL	Full scale select for accelerometer UI interface output 000: $\pm 32g$ (default) 001: $\pm 16g$ 010: $\pm 8g$ 011: $\pm 4g$ 100: Reserved 101: Reserved 110: Reserved 111: Reserved
4	-	Reserved
3:0	ACCEL_ODR	Accelerometer ODR selection for UI interface output 0000: Reserved 0001: 32kHz (LN mode) 0010: 16kHz (LN mode) 0011: 8kHz (LN mode) 0100: 4kHz (LN mode) 0101: 2kHz (LN mode) 0110: 1kHz (LN mode) (default) 0111: 200Hz (LP or LN mode) 1000: 100Hz (LP or LN mode) 1001: 50Hz (LP or LN mode) 1010: 25Hz (LP or LN mode) 1011: 12.5Hz (LP or LN mode) 1100: 6.25Hz (LP mode) 1101: 3.125Hz (LP mode) 1110: 1.5625Hz (LP mode) 1111: 500Hz (LP or LN mode)

13.32 GYRO_CONFIG1

Name: GYRO_CONFIG1
 Address: 81 (51h)
 Serial IF: R/W
 Reset value: 0x1A
 Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:5	TEMP_FILT_BW	Sets the bandwidth of the temperature signal DLPF 000: DLPF BW = 4000Hz; DLPF Latency = 0.125ms (default) 001: DLPF BW = 170Hz; DLPF Latency = 1ms 010: DLPF BW = 82Hz; DLPF Latency = 2ms 011: DLPF BW = 40Hz; DLPF Latency = 4ms 100: DLPF BW = 20Hz; DLPF Latency = 8ms 101: DLPF BW = 10Hz; DLPF Latency = 16ms 110: DLPF BW = 5Hz; DLPF Latency = 32ms 111: DLPF BW = 5Hz; DLPF Latency = 32ms
4	-	Reserved
3:2	GYRO_UI_FILT_ORD	Selects order of GYRO UI filter 00: 1 st Order 01: 2 nd Order 10: 3 rd Order 11: Reserved
1:0	GYRO_DEC2_M2_ORD	Selects order of GYRO DEC2_M2 Filter 00: Reserved 01: Reserved 10: 3 rd Order 11: Reserved

13.33 GYRO_ACCEL_CONFIG0

Name: GYRO_ACCEL_CONFIG0

Address: 82 (52h)

Serial IF: R/W

Reset value: 0x11

Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:4	ACCEL_UI_FILT_BW	LN Mode: Bandwidth for Accel LPF 0 BW=ODR/2 1 BW=max(400Hz, ODR)/4 (default) 2 BW=max(400Hz, ODR)/5 3 BW=max(400Hz, ODR)/8 4 BW=max(400Hz, ODR)/10 5 BW=max(400Hz, ODR)/16 6 BW=max(400Hz, ODR)/20 7 BW=max(400Hz, ODR)/40 8 to 13: Reserved 14 Low Latency option: Trivial decimation @ ODR of Dec2 filter output. Dec2 runs at max(400Hz, ODR) 15 Low Latency option: Trivial decimation @ ODR of Dec2 filter output. Dec2 runs at max(200Hz, 8*ODR) LP Mode: 0 Reserved 1 1x AVG filter (default) 2 to 5 Reserved 6 16x AVG filter 7 to 15 Reserved
3:0	GYRO_UI_FILT_BW	LN Mode: Bandwidth for Gyro LPF 0 BW=ODR/2 1 BW=max(400Hz, ODR)/4 (default) 2 BW=max(400Hz, ODR)/5 3 BW=max(400Hz, ODR)/8 4 BW=max(400Hz, ODR)/10 5 BW=max(400Hz, ODR)/16 6 BW=max(400Hz, ODR)/20 7 BW=max(400Hz, ODR)/40 8 to 13: Reserved 14 Low Latency option: Trivial decimation @ ODR of Dec2 filter output. Dec2 runs at max(400Hz, ODR) 15 Low Latency option: Trivial decimation @ ODR of Dec2 filter output. Dec2 runs at max(200Hz, 8*ODR)

13.34 ACCEL_CONFIG1

Name: ACCEL_CONFIG1 Address: 83 (53h) Serial IF: R/W Reset value: 0x15 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:5	-	Reserved
4:3	ACCEL_UI_FILT_ORD	Selects order of ACCEL UI filter 00: 1 st Order 01: 2 nd Order 10: 3 rd Order 11: Reserved
2:1	ACCEL_DEC2_M2_ORD	Order of Accelerometer DEC2_M2 filter 00: Reserved 01: Reserved 10: 3 rd order 11: Reserved
0	-	Reserved

13.35 TMST_CONFIG

Name: TMST_CONFIG Address: 84 (54h) Serial IF: R/W Reset value: 0x20 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:5	-	Reserved
4	TMST_TO_REGS_EN	0: TMST_VALUE[19:0] read always returns 0s 1: TMST_VALUE[19:0] read returns timestamp value
3	TMST_RES	Time Stamp resolution: When set to 0 (default), time stamp resolution is 1 μ s. When set to 1, resolution is 16 μ s
2	TMST_DELTA_EN	Time Stamp delta enable: When set to 1, the time stamp field contains the measurement of time since the last occurrence of ODR.
1	TMST_FSYNC_EN	Time Stamp register FSYNC enable (default). When set to 1, the contents of the Timestamp feature of FSYNC is enabled. The user also needs to select FIFO_TMST_FSYNC_EN in order to propagate the timestamp value to the FIFO.
0	TMST_EN	0: Time Stamp register disable 1: Time Stamp register enable (default)

13.36 WOM_CONFIG

Name: WOM_CONFIG Address: 87 (57h) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:4	-	Reserved
3	WOM_INT_MODE	0: Set WoM interrupt on the OR of all enabled accelerometer thresholds 1: Set WoM interrupt on the AND of all enabled accelerometer threshold
2	WOM_MODE	0: Initial sample is stored. Future samples are compared to initial sample 1: Compare current sample to previous sample
1:0	WOM_MODE1	00: Reserved 01: Set to 1 for WOM interrupt configuration 10: Reserved 11: Reserved

13.37 FIFO_CONFIG1

Name: FIFO_CONFIG1 Address: 95 (5Fh) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7	-	Reserved
6	FIFO_RESUME_PARTIAL_RD	0: Partial FIFO read disabled, requires re-reading of the entire FIFO 1: FIFO read can be partial, and resume from last read point
5	FIFO_WM_GT_TH	Trigger FIFO watermark interrupt on every ODR (DMA write) if $FIFO_COUNT \geq FIFO_WM_TH$
4	-	Reserved
3	FIFO_TMST_FSYNC_EN	Must be set to 1 for all FIFO use cases when FSYNC is used
2	FIFO_TEMP_EN	Enable temperature sensor packets to go to FIFO
1	FIFO_GYRO_EN	Enable gyroscope packets to go to FIFO
0	FIFO_ACCEL_EN	Enable accelerometer packets to go to FIFO

13.38 FIFO_CONFIG2

Name: FIFO_CONFIG2 Address: 96 (60h) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	FIFO_WM[7:0]	Lower bits of FIFO watermark. Generate interrupt when the FIFO reaches or exceeds FIFO_WM size in bytes or records according to FIFO_COUNT_REC setting. FIFO_WM_EN must be zero before writing this register. Interrupt only fires once. This register should be set to non-zero value, before choosing this interrupt source.

13.39 FIFO_CONFIG3

Name: FIFO_CONFIG3 Address: 97 (61h) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:4	-	Reserved
3:0	FIFO_WM[11:8]	Upper bits of FIFO watermark. Generate interrupt when the FIFO reaches or exceeds FIFO_WM size in bytes or records according to FIFO_COUNT_REC setting. FIFO_WM_EN must be zero before writing this register. Interrupt only fires once. This register should be set to non-zero value, before choosing this interrupt source.

Note: Do not set FIFO_WM to value 0.

13.40 FSYNC_CONFIG

Name: FSYNC_CONFIG Address: 98 (62h) Serial IF: R/W Reset value: 0x10 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7	-	Reserved
6:4	FSYNC_UI_SEL	000: Do not tag FSYNC flag 001: Tag FSYNC flag to TEMP_OUT LSB 010: Tag FSYNC flag to GYRO_XOUT LSB 011: Tag FSYNC flag to GYRO_YOUT LSB 100: Tag FSYNC flag to GYRO_ZOUT LSB 101: Tag FSYNC flag to ACCEL_XOUT LSB 110: Tag FSYNC flag to ACCEL_YOUT LSB 111: Tag FSYNC flag to ACCEL_ZOUT LSB
3:2	-	Reserved
1	FSYNC_UI_FLAG_CLEAR_SEL	0: FSYNC flag is cleared when UI sensor register is updated 1: FSYNC flag is cleared when UI interface reads the sensor register LSB of FSYNC tagged axis
0	FSYNC_POLARITY	0: Start from Rising edge of FSYNC pulse to measure FSYNC interval 1: Start from Falling edge of FSYNC pulse to measure FSYNC interval

13.41 INT_CONFIG0

Name: INT_CONFIG0 Address: 99 (63h) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:6	-	Reserved
5:4	UI_DRDY_INT_CLEAR	Data Ready Interrupt Clear Option (latched mode) 00: Clear on Status Bit Read (default) 01: Clear on Status Bit Read 10: Clear on FIFO data 1Byte Read 11: Clear on Status Bit Read AND on FIFO data 1 byte read
3:2	FIFO_THS_INT_CLEAR	FIFO Threshold Interrupt Clear Option (latched mode) 00: Clear on Status Bit Read (default) 01: Clear on Status Bit Read 10: Clear on FIFO data 1Byte Read 11: Clear on Status Bit Read AND on FIFO data 1 byte read
1:0	FIFO_FULL_INT_CLEAR	FIFO Full Interrupt Clear Option (latched mode) 00: Clear on Status Bit Read (default) 01: Clear on Status Bit Read 10: Clear on FIFO data 1Byte Read 11: Clear on Status Bit Read AND on FIFO data 1 byte read

13.42 INT_CONFIG1

Name: INT_CONFIG1 Address: 100 (64h) Serial IF: R/W Reset value: 0x10 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7	-	Reserved
6	INT_TPULSE_DURATION	Interrupt pulse duration 0: Interrupt pulse duration is 100μs. Use only if ODR < 4kHz. (Default) 1: Interrupt pulse duration is 8 μs. Required if ODR ≥ 4kHz, optional for ODR < 4kHz.
5	INT_TDEASSERT_DISABLE	Interrupt de-assertion duration 0: The interrupt de-assertion duration is set to a minimum of 100μs. Use only if ODR < 4kHz. (Default) 1: Disables de-assert duration. Required if ODR ≥ 4kHz, optional for ODR < 4kHz.
4	INT_ASYNC_RESET	User should change setting to 0 from default setting of 1, for proper INT1 and INT2 pin operation
3:0	-	Reserved

13.43 INT_SOURCE0

Name: INT_SOURCE0 Address: 101 (65h) Serial IF: R/W Reset value: 0x10 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7	-	Reserved
6	UI_FSYNC_INT1_EN	0: UI FSYNC interrupt not routed to INT1 1: UI FSYNC interrupt routed to INT1
5	PLL_RDY_INT1_EN	0: PLL ready interrupt not routed to INT1 1: PLL ready interrupt routed to INT1
4	RESET_DONE_INT1_EN	0: Reset done interrupt not routed to INT1 1: Reset done interrupt routed to INT1
3	UI_DRDY_INT1_EN	0: UI data ready interrupt not routed to INT1 1: UI data ready interrupt routed to INT1
2	FIFO_THS_INT1_EN	0: FIFO threshold interrupt not routed to INT1 1: FIFO threshold interrupt routed to INT1
1	FIFO_FULL_INT1_EN	0: FIFO full interrupt not routed to INT1 1: FIFO full interrupt routed to INT1
0	UI_AGC_RDY_INT1_EN	0: UI AGC ready interrupt not routed to INT1 1: UI AGC ready interrupt routed to INT1

13.44 INT_SOURCE1

Name: INT_SOURCE1 Address: 102 (66h) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:3	-	Reserved
2	WOM_Z_INT1_EN	0: Z-axis WOM interrupt not routed to INT1 1: Z-axis WOM interrupt routed to INT1
1	WOM_Y_INT1_EN	0: Y-axis WOM interrupt not routed to INT1 1: Y-axis WOM interrupt routed to INT1
0	WOM_X_INT1_EN	0: X-axis WOM interrupt not routed to INT1 1: X-axis WOM interrupt routed to INT1

13.45 INT_SOURCE3

Name: INT_SOURCE3 Address: 104 (68h) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7	-	Reserved
6	UI_FSYNC_INT2_EN	0: UI FSYNC interrupt not routed to INT2 1: UI FSYNC interrupt routed to INT2
5	PLL_RDY_INT2_EN	0: PLL ready interrupt not routed to INT2 1: PLL ready interrupt routed to INT2
4	RESET_DONE_INT2_EN	0: Reset done interrupt not routed to INT2 1: Reset done interrupt routed to INT2
3	UI_DRDY_INT2_EN	0: UI data ready interrupt not routed to INT2 1: UI data ready interrupt routed to INT2
2	FIFO_THS_INT2_EN	0: FIFO threshold interrupt not routed to INT2 1: FIFO threshold interrupt routed to INT2
1	FIFO_FULL_INT2_EN	0: FIFO full interrupt not routed to INT2 1: FIFO full interrupt routed to INT2
0	UI_AGC_RDY_INT2_EN	0: UI AGC ready interrupt not routed to INT2 1: UI AGC ready interrupt routed to INT2

13.46 INT_SOURCE4

Name: INT_SOURCE4 Address: 105 (69h) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:3	-	Reserved
2	WOM_Z_INT2_EN	0: Z-axis WOM interrupt not routed to INT2 1: Z-axis WOM interrupt routed to INT2
1	WOM_Y_INT2_EN	0: Y-axis WOM interrupt not routed to INT2 1: Y-axis WOM interrupt routed to INT2
0	WOM_X_INT2_EN	0: X-axis WOM interrupt not routed to INT2 1: X-axis WOM interrupt routed to INT2

13.47 FIFO_LOST_PKT0

Name: FIFO_LOST_PKT0 Address: 108 (6Ch) Serial IF: R Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	FIFO_LOST_PKT_CNT[7:0]	Low byte, number of packets lost in the FIFO

13.48 FIFO_LOST_PKT1

Name: FIFO_LOST_PKT1
Address: 109 (6Dh)
Serial IF: R
Reset value: 0x00
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	FIFO_LOST_PKT_CNT[15:8]	High byte, number of packets lost in the FIFO

13.49 SELF_TEST_CONFIG

Name: SELF_TEST_CONFIG
Address: 112 (70h)
Serial IF: R/W
Reset value: 0x00
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7	-	Reserved
6	ACCEL_ST_POWER	Set to 1 for accel self-test Otherwise set to 0; Set to 0 after self-test is completed
5	EN_AZ_ST	Enable Z-accel self-test
4	EN_AY_ST	Enable Y-accel self-test
3	EN_AX_ST	Enable X-accel self-test
2	EN_GZ_ST	Enable Z-gyro self-test
1	EN_GY_ST	Enable Y-gyro self-test
0	EN_GX_ST	Enable X-gyro self-test

13.50 WHO_AM_I

Name: WHO_AM_I
Address: 117 (75h)
Serial IF: R
Reset value: 0x3B
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	WHOAMI	Register to indicate to user which device is being accessed

Description:

This register is used to verify the identity of the device. The contents of WHOAMI is an 8-bit device ID. The default value of the register is 0x3B. This is different from the I²C address of the device as seen on the slave I²C controller by the applications processor.

13.51 REG_BANK_SEL

Note: This register is accessible from all register banks

Name: REG_BANK_SEL Address: 118 (76h) Serial IF: R/W Reset value: 0x00 Clock Domain: ALL		
BIT	NAME	FUNCTION
7:3	-	Reserved
2:0	BANK_SEL	Register bank selection 000: Bank 0 (default) 001: Bank 1 010: Bank 2 011: Bank 3 100: Bank 4 101: Reserved 110: Reserved 111: Reserved

14 USER BANK 1 REGISTER MAP – DESCRIPTIONS

This section describes the function and contents of each register within USR Bank 1.

14.1 SENSOR_CONFIG0

Name: SENSOR_CONFIG0 Address: 03 (03h) Serial IF: R/W Reset value: 0x40 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:6	-	Reserved
5	ZG_DISABLE	0: Z gyroscope is on 1: Z gyroscope is disabled
4	YG_DISABLE	0: Y gyroscope is on 1: Y gyroscope is disabled
3	XG_DISABLE	0: X gyroscope is on 1: X gyroscope is disabled
2	ZA_DISABLE	0: Z accelerometer is on 1: Z accelerometer is disabled
1	YA_DISABLE	0: Y accelerometer is on 1: Y accelerometer is disabled
0	XA_DISABLE	0: X accelerometer is on 1: X accelerometer is disabled

14.2 GYRO_CONFIG_STATIC2

Name: GYRO_CONFIG_STATIC2 Address: 11 (0Bh) Serial IF: R/W Reset value: 0x03 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:2	-	Reserved
1	GYRO_AAF_DIS	0: Enable Anti-Aliasing/Low Pass Filter 1: Disable Anti-Aliasing/Low Pass Filter (default)
0	GYRO_NF_DIS	0: Enable Notch Filter 1: Disable Notch Filter (default)

14.3 GYRO_CONFIG_STATIC3

Name: GYRO_CONFIG_STATIC3 Address: 12 (0Ch) Serial IF: R/W Reset value: 0x3F Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:6	-	Reserved
5:0	GYRO_AAF_DELT	Controls bandwidth of the gyroscope anti-alias filter See section 5.2 for details

14.4 GYRO_CONFIG_STATIC4

Name: GYRO_CONFIG_STATIC4 Address: 13 (0Dh) Serial IF: R/W Reset value: 0x81 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	GYRO_AAF_DELTSQR[7:0]	Controls bandwidth of the gyroscope anti-alias filter See section 5.2 for details

14.5 GYRO_CONFIG_STATIC5

Name: GYRO_CONFIG_STATIC5 Address: 14 (0Eh) Serial IF: R/W Reset value: 0x3F Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:4	GYRO_AAF_BITSHIFT	Controls bandwidth of the gyroscope anti-alias filter See section 5.2 for details
3:0	GYRO_AAF_DELTSQR[11:8]	Controls bandwidth of the gyroscope anti-alias filter See section 5.2 for details

14.6 GYRO_CONFIG_STATIC6

Name: GYRO_CONFIG_STATIC6 Address: 15 (0Fh) Serial IF: R/W Reset value: 0xEA Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	GYRO_X_NF_COSWZ[7:0]	Used for gyroscope X-axis notch filter frequency selection See section 5.1 for details

14.7 GYRO_CONFIG_STATIC7

Name: GYRO_CONFIG_STATIC7 Address: 16 (10h) Serial IF: R/W Reset value: 0x28 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	GYRO_Y_NF_COSWZ[7:0]	Used for gyroscope Y-axis notch filter frequency selection See section 5.1 for details

14.8 GYRO_CONFIG_STATIC8

Name: GYRO_CONFIG_STATIC8 Address: 17 (11h) Serial IF: R/W Reset value: 0x07 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	GYRO_Z_NF_COSWZ[7:0]	Used for gyroscope Z-axis notch filter frequency selection See section 5.1 for details

14.9 GYRO_CONFIG_STATIC9

Name: GYRO_CONFIG_STATIC9 Address: 18 (12h) Serial IF: R/W Reset value: 0x01 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:6	-	Reserved
5	GYRO_Z_NF_COSWZ_SEL[0]	Used for gyroscope Z-axis notch filter frequency selection See section 5.1 for details
4	GYRO_Y_NF_COSWZ_SEL[0]	Used for gyroscope Y-axis notch filter frequency selection See section 5.1 for details
3	GYRO_X_NF_COSWZ_SEL[0]	Used for gyroscope X-axis notch filter frequency selection See section 5.1 for details
2	GYRO_Z_NF_COSWZ[8]	Used for gyroscope Z-axis notch filter frequency selection See section 5.1 for details
1	GYRO_Y_NF_COSWZ[8]	Used for gyroscope Y-axis notch filter frequency selection See section 5.1 for details
0	GYRO_X_NF_COSWZ[8]	Used for gyroscope X-axis notch filter frequency selection See section 5.1 for details

14.10 GYRO_CONFIG_STATIC10

Name: GYRO_CONFIG_STATIC10 Address: 19 (13h) Serial IF: R/W Reset value: 0x11 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7	-	Reserved
6:4	GYRO_NF_BW_SEL	Selects bandwidth for gyroscope notch filter See section 5.1 for details
3:1	GYRO_HPF_BW_IND	Selects HPF 3dB cutoff frequency bandwidth See section 5.6 for details
0	GYRO_HPF_ORD_IND	Selects HPF filter order (see section 5.6 for details) 0: 1 st order HPF 1: 2 nd order HPF

14.11 XG_ST_DATA

Name: XG_ST_DATA
 Address: 95 (5Fh)
 Serial IF: R/W
 Reset value: 0xXX (The value in this register indicates the self-test output generated during manufacturing tests)
 Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	XG_ST_DATA	X-gyro self-test data

14.12 YG_ST_DATA

Name: YG_ST_DATA
 Address: 96 (60h)
 Serial IF: R/W
 Reset value: 0xXX (The value in this register indicates the self-test output generated during manufacturing tests)
 Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	YG_ST_DATA	Y-gyro self-test data

14.13 ZG_ST_DATA

Name: ZG_ST_DATA
 Address: 97 (61h)
 Serial IF: R/W
 Reset value: 0xXX (The value in this register indicates the self-test output generated during manufacturing tests)
 Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	ZG_ST_DATA	Z-gyro self-test data

14.14 TMSTVAL0

Name: TMSTVAL0
 Address: 98 (62h)
 Serial IF: R
 Reset value: 0x00
 Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	TMST_VALUE[7:0]	When TMST_STROBE is programmed, the current value of the internal counter is latched to this register. Allows the full 20-bit precision of the time stamp to be read back.

14.15 TMSTVAL1

Name: TMSTVAL1
 Address: 99 (63h)
 Serial IF: R
 Reset value: 0x00
 Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	TMST_VALUE[15:8]	When TMST_STROBE is programmed, the current value of the internal counter is latched to this register. Allows the full 20-bit precision of the time stamp to be read back.

14.16 TMSTVAL2

Name: TMSTVAL2 Address: 100 (64h) Serial IF: R Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:4	-	Reserved
3:0	TMST_VALUE[19:16]	When TMST_STROBE is programmed, the current value of the internal counter is latched to this register. Allows the full 20-bit precision of the time stamp to be read back.

14.17 INTF_CONFIG4

Name: INTF_CONFIG4 Address: 122 (7Ah) Serial IF: R/W Reset value: 0x02 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:2	-	Reserved
1	SPI_AP_4WIRE	0: AP interface uses 3-wire SPI mode 1: AP interface uses 4-wire SPI mode (default)
0	-	Reserved

14.18 INTF_CONFIG5

Name: INTF_CONFIG5 Address: 123 (7Bh) Serial IF: R/W Reset value: 0x24 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:3	-	Reserved
2:1	PIN19_FUNCTION	Selects among the following functionalities for pin 9 00: INT2 01: FSYNC 10: Reserved 11: Reserved
0	-	Reserved

15 USER BANK 2 REGISTER MAP – DESCRIPTIONS

This section describes the function and contents of each register within USR Bank 2.

15.1 ACCEL_CONFIG_STATIC2

Name: ACCEL_CONFIG_STATIC2 Address: 03 (03h) Serial IF: R/W Reset value: 0x7F Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7	-	Reserved
6:1	ACCEL_AAF_DELT	Controls bandwidth of the accelerometer anti-alias filter See section 5.2 for details
0	ACCEL_AAF_DIS	0: Enable accelerometer anti-aliasing filter 1: Disable accelerometer anti-aliasing filter (default)

15.2 ACCEL_CONFIG_STATIC3

Name: ACCEL_CONFIG_STATIC3 Address: 04 (04h) Serial IF: R/W Reset value: 0x81 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	ACCEL_AAF_DELTSQR[7:0]	Controls bandwidth of the accelerometer anti-alias filter See section 5.2 for details

15.3 ACCEL_CONFIG_STATIC4

Name: ACCEL_CONFIG_STATIC4 Address: 05 (05h) Serial IF: R/W Reset value: 0x3F Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:4	ACCEL_AAF_BITSHIFT	Controls bandwidth of the accelerometer anti-alias filter See section 5.2 for details
3:0	ACCEL_AAF_DELTSQR[11:8]	Controls bandwidth of the accelerometer anti-alias filter See section 5.2 for details

15.4 XA_ST_DATA

Name: XA_ST_DATA Address: 59 (3Bh) Serial IF: R/W Reset value: 0xXX (The value in this register indicates the self-test output generated during manufacturing tests) Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	XA_ST_DATA	X-accel self-test data

15.5 YA_ST_DATA

Name: YA_ST_DATA
Address: 60 (3Ch)
Serial IF: R/W
Reset value: 0xXX (The value in this register indicates the self-test output generated during manufacturing tests)
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	YA_ST_DATA	Y-accel self-test data

15.6 ZA_ST_DATA

Name: ZA_ST_DATA
Address: 61 (3Dh)
Serial IF: R/W
Reset value: 0xXX (The value in this register indicates the self-test output generated during manufacturing tests)
Clock Domain: SCLK_UI

BIT	NAME	FUNCTION
7:0	ZA_ST_DATA	Z-accel self-test data

16 USER BANK 4 REGISTER MAP – DESCRIPTIONS

This section describes the function and contents of each register within USB Bank 4.

16.1 ACCEL_WOM_X_THR

Name: ACCEL_WOM_X_THR Address: 74 (4Ah) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	WOM_X_TH	Threshold value for the Wake on Motion Interrupt for X-axis accelerometer WoM thresholds are expressed in fixed “mg” independent of the selected Range [0g : 1g]; Resolution 1g/256≈3.9mg

16.2 ACCEL_WOM_Y_THR

Name: ACCEL_WOM_Y_THR Address: 75 (4Bh) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	WOM_Y_TH	Threshold value for the Wake on Motion Interrupt for Y-axis accelerometer WoM thresholds are expressed in fixed “mg” independent of the selected Range [0g : 1g]; Resolution 1g/256≈3.9mg

16.3 ACCEL_WOM_Z_THR

Name: ACCEL_WOM_Z_THR Address: 76 (4Ch) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	WOM_Z_TH	Threshold value for the Wake on Motion Interrupt for Z-axis accelerometer WoM thresholds are expressed in fixed “mg” independent of the selected Range [0g : 1g]; Resolution 1g/256≈3.9mg

16.4 OFFSET_USER0

Name: OFFSET_USER0 Address: 119 (77h) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	GYRO_X_OFFUSER[7:0]	Lower bits of X-gyro offset programmed by user. Max value is ±64 dps, resolution is 1/32 dps.

16.5 OFFSET_USER1

Name: OFFSET_USER1 Address: 120 (78h) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:4	GYRO_Y_OFFUSER[11:8]	Upper bits of Y-gyro offset programmed by user. Max value is ± 64 dps, resolution is 1/32 dps.
3:0	GYRO_X_OFFUSER[11:8]	Upper bits of X-gyro offset programmed by user. Max value is ± 64 dps, resolution is 1/32 dps.

16.6 OFFSET_USER2

Name: OFFSET_USER2 Address: 121 (79h) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	GYRO_Y_OFFUSER[7:0]	Lower bits of Y-gyro offset programmed by user. Max value is ± 64 dps, resolution is 1/32 dps.

16.7 OFFSET_USER3

Name: OFFSET_USER3 Address: 122 (7Ah) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	GYRO_Z_OFFUSER[7:0]	Lower bits of Z-gyro offset programmed by user. Max value is ± 64 dps, resolution is 1/32 dps.

16.8 OFFSET_USER4

Name: OFFSET_USER4 Address: 123 (7Bh) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:4	ACCEL_X_OFFUSER[11:8]	Upper bits of X-accel offset programmed by user. Max value is $\pm 1g$, resolution is 0.5mg.
3:0	GYRO_Z_OFFUSER[11:8]	Upper bits of Z-gyro offset programmed by user. Max value is ± 64 dps, resolution is 1/32 dps.

16.9 OFFSET_USER5

Name: OFFSET_USER5 Address: 124 (7Ch) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	ACCEL_X_OFFUSER[7:0]	Lower bits of X-accel offset programmed by user. Max value is $\pm 1g$, resolution is 0.5mg.

16.10 OFFSET_USER6

Name: OFFSET_USER6 Address: 125 (7Dh) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	ACCEL_Y_OFFUSER[7:0]	Lower bits of Y-accel offset programmed by user. Max value is $\pm 1g$, resolution is 0.5mg.

16.11 OFFSET_USER7

Name: OFFSET_USER7 Address: 126 (7Eh) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:4	ACCEL_Z_OFFUSER[11:8]	Upper bits of Z-accel offset programmed by user. Max value is $\pm 1g$, resolution is 0.5mg.
3:0	ACCEL_Y_OFFUSER[11:8]	Upper bits of Y-accel offset programmed by user. Max value is $\pm 1g$, resolution is 0.5mg.

16.12 OFFSET_USER8

Name: OFFSET_USER8 Address: 127 (7Fh) Serial IF: R/W Reset value: 0x00 Clock Domain: SCLK_UI		
BIT	NAME	FUNCTION
7:0	ACCEL_Z_OFFUSER[7:0]	Lower bits of Z-accel offset programmed by user. Max value is $\pm 1g$, resolution is 0.5mg.

17 REFERENCE

Please refer to “InvenSense MEMS Handling Application Note (AN-IVS-0002A-00)” for the following information:

- Manufacturing Recommendations
 - Assembly Guidelines and Recommendations
 - PCB Design Guidelines and Recommendations
 - MEMS Handling Instructions
 - ESD Considerations
 - Reflow Specification
 - Storage Specifications
 - Package Marking Specification
 - Tape & Reel Specification
 - Reel & Pizza Box Label
 - Packaging
 - Representative Shipping Carton Label
- Compliance
 - Environmental Compliance
 - DRC Compliance
 - Compliance Declaration Disclaimer

18 REVISION HISTORY

Revision Date	Revision	Description
08/01/2019	1.0	Initial Release
10/24/2019	1.1	Updated Section 9
06/07/2022	1.2	Formatting updates

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